

Monotone and oscillatory traveling waves for Burgers equations with time-delayed flux and singular fast diffusion

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Abstract

In this paper, we first derive a new model from nonlocal population dynamics with age-structure, which is presented in the form of Burgers equation with time-delayed flux and nonlinear diffusion: $u_t - D(u^{m-1}u_x)_x + f(u(t-r, x))_x = 0$. Here, $r > 0$ is the time-delay, the matured age of single species, m is the index of nonlinear diffusion. When $m < 1$, the nonlinear diffusion reduces to the fast diffusion $(\frac{u_x}{u^{1-m}})_x$ with singularity at $u = 0$. The main issue of the paper is to show the criteria for the existence of traveling waves and their monotonicity and oscillation. The flux function and the time-delay play the crucial roles for the existence of traveling waves and their monotonicity and oscillations, which are completely different from the existing studies. When the time-delay is big, the expected traveling waves lose their monotonicity under the bad influence of the time delay, and behave as oscillating waves, which introduces significant challenges for establishing existence. To overcome the difficulty, we define several special spaces which are related to the asymptotic behavior in the far fields $\pm\infty$ and their oscillations, and utilize the Leray-Schauder theorem to prove the existence of oscillatory traveling wave solutions. Finally, some numerical simulations are carried out, which perfectly demonstrate all types of monotone and oscillatory traveling waves.

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1 Introduction

New Model: Burgers equation with time-delayed flux. Concerning with the population of single species, Metz and Diekmann [39] in 1986 first proposed a standard argument on population with age structure and diffusion:

$$\frac{\partial v}{\partial t} + \frac{\partial v}{\partial a} = D(a) \frac{\partial^2 v}{\partial x^2} - d(a)v, \quad (1.1)$$

where $v(t, a, x)$ denotes the density of population at time $t \geq 0$, age $a \geq 0$ and location x in the domain $\Omega = (-\infty, \infty)$, $D(a)$ and $d(a)$ are the spatial diffusion rate and death rate, respectively, at age a . Let $r > 0$ be the maturation age for the species. Then the total matured population at time t and location x is given by

$$u(t, x) = \int_r^\infty v(t, a, x) da.$$

After some structural setting for simplicity, by using Fourier transform, So, Wu and Zou [46] derived the dynamical equation for the total matured population $u(t, x)$, the nonlocal reaction-diffusion equations with time delay as follows, see also [37, 38]

$$u_t - Du_{xx} + d(u) = \int_{\mathbb{R}} K(x - y)b(u(t - r, y)) dy, \quad (1.2)$$

where $D > 0$ is the constant diffusion coefficient, $b(u(t - r, x))$ is the birth rate function, $r > 0$ is the time delay, $d(u)$ is the death rate function, and the kernel $K(x)$ is positive and unit

$$K(x) > 0, \quad \int_{\mathbb{R}} K(y) dy = 1. \quad (1.3)$$

Regarding the case of single population's flocking migration, such as jackdaws and locusts moving in group, the above linear diffusion equation (1.2) then can be modified as the density-dependent diffusion:

$$u_t - D(Q(u))_{xx} + d(u) = \int_{\mathbb{R}} K(x - y)b(u(t - r, y)) dy, \quad (1.4)$$

where $Q(u) \geq 0$ is the group density for population's flocking migration.

Formally, by Taylor's expansion and using the unit property of kernel $K(y)$ in (1.3), we have

$$\begin{aligned} & \int_{\mathbb{R}} K(y)b(u(t - r, x - y)) dy - b(u(t - r, x)) \\ &= \int_{\mathbb{R}} K(y) [b(u(t - r, x - y)) - b(u(t - r, x))] dy \\ &= - \left(\int_{\mathbb{R}} K(y)y dy \right) b(u(t - r, x))_x + \left(\frac{1}{2} \int_{\mathbb{R}} K(y)y^2 dy \right) b(u(t - r, x))_{xx} \\ & \quad + o(1)b(u(t - r, x))_{xx} \int_{\mathbb{R}} K(y)y^2 dy \\ &\approx -Ab(u(t - r, x))_x + Bb(u(t - r, x))_{xx}, \end{aligned}$$

where $A = \int_{\mathbb{R}} K(y)y dy$ and $B = \int_{\mathbb{R}} K(y)y^2 dy$. When the kernel $K(y)$ is non-symmetric:

$$K(-y) \neq K(y),$$

then $A = \int_{\mathbb{R}} K(y)dy \neq 0$. So we roughly reduce the nonlocal reaction-diffusion equation of population dynamics to an advection-reaction-diffusion equation with time-delay:

$$u_t - D(Q(u))_{xx} + d(u) + Ab(u(t-r, x))_x = Bb(u(t-r, x))_{xx} + b(u(t-r, x)),$$

which can be rewritten in a more general form

$$u_t + f(u(t-r, x))_x - D(Q(u))_{xx} - g(u(t-r, x))_{xx} + d(u) = b(u(t-r, x)), \quad (1.5)$$

for some more general functions $f(u)$, $D(u) \geq 0$ with $D'(u) \geq 0$, $g(u)$, $d(u)$ and $b(u)$ to be given.

Furthermore, for simplicity this equation can be reduced, by taking $f = g = 0$, to the following time-delayed reaction-diffusion equation:

$$u_t - D(Q(u))_{xx} + d(u) = b(u(t-r, x)), \quad (1.6)$$

which was well-studied in [18, 54, 55].

By taking $d = g = b = 0$, then we derive the following *time-delayed Burgers equation* from (1.5), a first model for Burgers equations involving time-delay in flux,

$$u_t + f(u(t-r, x))_x - D(Q(u))_{xx} = 0. \quad (1.7)$$

When we take $Q(u) = u^m$, the above time-delayed Burgers equation with nonlinear diffusion (1.7) is then reduced to the classic porous media type diffusion:

$$u_t + f(u(t-r, x))_x - D(u^m)_{xx} = 0. \quad (1.8)$$

To study this new model will be our main focus in the present paper.

Regarding the porous media type diffusion (or say, viscosity in fluid dynamics): $(u^m)_{xx} = m(u^{m-1}u_x)_x$, in order to avoid the unnecessary confusions caused by the index with $m < 0$ and $m = 0$, the equation (1.8) can be essentially written as

$$u_t + f(u(t-r, x))_x - D(u^{m-1}u_x)_x = 0. \quad (1.9)$$

Based on different value of m , the above equation (1.9) can be classified as the degenerate (slow) diffusion equation with $m > 1$:

$$u_t + f(u(t-r, x))_x - D(u^{m-1}u_x)_x = 0, \quad \text{for } m > 1, \quad (1.10)$$

whose solution usually losses its regularity and becomes non-differentiable. Sharp traveling waves arise in this case, which will be considered in future as the second part of this series studies.

While, when $m = 1$, the equation (1.9) is with the regular diffusion:

$$u_t + f(u(t-r, x))_x - Du_{xx} = 0, \quad \text{for } m = 1, \quad (1.11)$$

When $m < 1$, the equation (1.9) is called the fast diffusion equation with singularity at $u = 0$:

$$u_t + f(u(t-r, x))_x - D\left(\frac{u_x}{u^{1-m}}\right)_x = 0, \quad \text{for } m < 1. \quad (1.12)$$

In particular, when $0 < m < 1$, it is called as regular fast diffusion:

$$u_t + f(u(t-r, x))_x - D\left(\frac{u_x}{u^{1-m}}\right)_x = 0, \quad \text{for } 0 < m < 1, \quad (1.13)$$

and when $m = 0$, it is called as critical fast diffusion:

$$u_t + f(u(t-r, x))_x - D(\ln u)_{xx} = 0, \quad \text{for } m = 0, \quad (1.14)$$

and when $m < 0$, it is called as super fast diffusion:

$$u_t + f(u(t-r, x))_x - D\left(\frac{u_x}{u^{1+|m|}}\right)_x = 0, \quad \text{for } m < 0. \quad (1.15)$$

In this paper, we are mainly interested in the following Burgers equations with time delay and fast diffusion:

$$0 < m < 1 : \begin{cases} u_t - D\left(\frac{u_x}{u^{1-m}}\right)_x + f(u(t-r, x))_x = 0, & x \in \mathbb{R}, t > 0, \\ u|_{t=\tau} = u_0(\tau, x), & x \in \mathbb{R}, \tau \in [-r, 0], \end{cases} \quad (1.16)$$

and

$$m = 0 : \begin{cases} u_t - D(\ln u)_{xx} + f(u(t-r, x))_x = 0, & x \in \mathbb{R}, t > 0, \\ u|_{t=\tau} = u_0(\tau, x), & x \in \mathbb{R}, \tau \in [-r, 0], \end{cases} \quad (1.17)$$

and

$$m < 0 : \begin{cases} u_t - D\left(\frac{u_x}{u^{1+|m|}}\right)_x + f(u(t-r, x))_x = 0, & x \in \mathbb{R}, t > 0, \\ u|_{t=\tau} = u_0(\tau, x), & x \in \mathbb{R}, \tau \in [-r, 0], \end{cases} \quad (1.18)$$

where f is the flux which is assumed to be suitably smooth, $r > 0$ is the time delay, and the initial value $u_0(\tau, x)$ satisfies

$$u_0(\tau, x) \rightarrow u_{\pm}, \quad (x \rightarrow \pm\infty).$$

The main issue of the paper is to look for the traveling wave solutions of (1.16), (1.17) and (1.18) connecting u_+ and u_- , where $u_{\pm}$ are state constants satisfying $0 = u_+ < u_-$.

For the above equations, different from the existing studies, there are two features: one is the time-delay which may reduce the solution to be oscillating; the other is the singular fast diffusion at $u_+ = 0$.

Background of studies. Tracing back to the pioneering work of Burgers [2, 3] on fluid turbulence, Burgers equation has emerged as a fundamental class of partial differential equations in the study of gas and fluid dynamics:

$$u_t + f(u)_x = Du_{xx}. \quad (1.19)$$

The study of (1.19) has yielded a great many interesting and important results. Among these, it is well known that viscous shocks and rarefaction waves are of particular significance, as they represent key nonlinear solutions to the equation. A viscous shock $\phi(x-st)$ of scalar viscous conservation laws (1.19) satisfying

$$\begin{cases} -s\phi' + f'(\phi)\phi' = D\phi'', \\ \phi(\pm\infty) = u_{\pm}, \end{cases} \quad (1.20)$$

where $u_- > u_+$ and s is the speed depended by the Rankine-Hugoniot condition

$$s = \frac{f(u_+) - f(u_-)}{u_+ - u_-}.$$

Through the phase plane analysis or the center manifold theorem demonstrated in the work of Kopell and Howard [25], the existence of travelling waves can be rigorously established. In 1960, Ili'in and Oleinik [19] first proved the stability of shock profiles with exponential decay for the inviscid conservation laws with convex nonlinearity by applying the maximum principle. After that, Sattinger [45] derived the stability through employing the spectral analysis. Later in 1985, Kawashima and Matsumura [22] utilized energy method to approach the asymptotic stability and convergence rate. In addition, based on the Hopf transformation and L^2 energy method, Nishihara [41] proved the stability of scalar viscous conservation laws. In the previous study, the flux $f(u)$ is convex, however, when $f(u)$ is nonconvex, a series of investigations had been conducted to explore the stability properties. The stability had been studied with the help of weighted energy estimate by Kawashima and Matsumura [23], which mentions the system case at the same time. Jones et al. [20] obtained the stability with algebraic time decay through spectral analysis approach, meanwhile, the convergence rates, in particular, the stability of degenerate viscous shock waves, were investigated by Matsumura and Nishihara [35] and Mei [36] respectively. Subsequently, Freistühler and Serre [7] acquired the L^1 stability of travelling waves for the scalar viscous conservation laws and the stability of algebraic decay with respect to time when the initial disturbance is arbitrarily large was given by Nishihara and Zhao [42]. Furthermore, Chung et al. [4] considered the following Burgers equation

$$\begin{aligned} u_t + uu_x - u_{xx} &= \delta(x), & x \in \mathbb{R}, t > 0, \\ u(x, 0) &= u_0(x), & x \in \mathbb{R}, \end{aligned}$$

where $\delta(x)$ is a Dirac delta measure, and the initial data u_0 is assumed to belong to the space $C^1(\mathbb{R}) \cap L^1(\mathbb{R}) \cap W^{1,\infty}(\mathbb{R})$. Based on the Cole-Hopf transformation and leveraging the properties of the initial data, the long time asymptotic convergence of the global weak solution to the equation have been demonstrated. Later, M. Sahoo et al. [44] derived the same asymptotic behavior without the assumption of initial data. It is worth mentioning the contributions of Weinberger [50], Howard [14, 15, 16], Engler [6], and [10, 12, 13, 24] have also provided significant insights into the multidimensional scalar case. Recently, Kang, Vasseur and Wang [21] proposed the relative entropy method and significantly improved the shock wave stability with large initial amplitude. Huang, Wang and Zhang [17] proved the stability of combination of viscous shock and rarefaction wave when the flux $f(u)$ is cubic. Xin, Yuan and Yuan [51] investigated the stability of viscous shock wave when the initial perturbation is with periodic oscillations.

For the system cases of conservation laws, the result of relatively early stability of shock profiles was investigated by Matsumura and Nishihara [34] and Goodman [9]. Subsequently, numerous studies have focused on the system case, including Liu [30], Liu and Zeng [31, 32], Szepessy and Xin [47], Matsumura and Mei [33], Wang, Yang and Zhao [49], Kang, Vasseur and Wang [21].

For the Burgers equation with time-delayed flux (1.8) without the time-delay, i.e., $r = 0$, then it is reduced to the standard nonlinear diffusion Burgers equation:

$$u_t + f(u)_x - D(u^{m-1}u_x)_x = 0, \quad (1.21)$$

which presents the model in fluid mechanics describing the interaction between nonlinear advection and nonlinear diffusion [26, 27], where the diffusion in this context is called “saturating dissipation” which is a nonlinear form of diffusion in which the rate of diffusion levels off or “saturates” as the solution steepens. Another example for the nonlinear diffusive Burgers equation is the spatially aggregating population model which provides the

homogenizing process due to density-dependent diffusion, and the homogenizing one due to a certain long-range transport, studied by Nagai and Mimura [40]. For fast diffusion with $m < 1$, it was used by Barenblatt, Zel'dovich and Kompaneets [1] to model gas-kinetics, then for thin liquid film dynamics with Van der Waals forces, and also for plasma (e.g. in nuclear reactors) when high temperatures affect standard physical laws (i.e. Fourier's Law); see for instance the monographs by Barenblatt [1] and Vázquez [48]. Additionally the case where plasma density satisfies fast-diffusion with $m = \frac{1}{2}$ was numerically studied by Drake, Greenwood, Navratil and Post [5] and Okuda and Dawson [43], respectively. Such a nonlinear diffusion Burgers equation (1.21) was also intensively discussed by Gilding and Kersner in their textbook [8].

In the context of the fast diffusion case ($m < 1$), the primary challenge arises from the singularity in the expressions $(\frac{u_x}{u^{1-m}})_x$ as u approaches $u_+ = 0$. Recently, when there is no time-delay ($r = 0$), the traveling waves for the equations (1.13)-(1.15) are monotone, and can be easily obtained by Mei and his collaborators [28, 29, 52] using the inverse function technique. They [28, 29, 52] further technically showed the stability of shock profiles with the singular fast diffusion by the weighted energy method, where the selection of appropriate singular-weight-functions plays a crucial role for overcoming the singularity in diffusion.

When $m > 1$, the equation (1.9) is degenerate in diffusion. Usually the solutions for such equations will lose their regularities and the traveling waves become the sharp waves, see the studies without time-delay in [56, 57, 53]. In this case, we expect the traveling waves to be sharp and oscillating when the time-delay is big. This will be our future target.

Difficulties and strategies. Completely different from the previous studies [28, 29, 52] for the singular fast diffusion (1.21) without time-delay, where the traveling waves (viscous shocks) are monotone, and converge exponentially to u_- in the far field when $f'(0) < s < f'(u_-)$, and converge algebraically to u_- when $f'(0) < s = f'(u_-)$, our investigations reveal that, even when $f'(0) < s = f'(u_-)$, the travelling waves still converge exponentially to u_- , the phenomenon caused by the introduction of time delay r . The most challenging issue concerning the existence of traveling wave solutions arises from the presence of time delay, where the presence of time delay implies that the traveling wave may not be monotonic, even they are oscillating. Thus, the inverse function technique used in [28, 29, 52] is not available anymore in our problems with time-delay. To overcome this difficulty, we first heuristically analyze the asymptotic (monotone or oscillatory) behaviors of the expected traveling waves, then we technically define some special solution spaces according to the asymptotic behavior at $\xi \rightarrow \pm\infty$ and apply the Leray-Schauder theorem in these spaces to address the impact brought about by time delay.

The structure of this paper is outlined as follows. In Section 2, we present the principal findings of our study, which encompass asymptotic behavior at $\xi \rightarrow \pm\infty$, oscillatory behavior, and existence of travelling waves. In Section 3, we perform formal asymptotic analysis of the travelling wave equations at both constant states $u = 0$ and $u = u_-$ to determine the expected asymptotic behavior and oscillation conditions. This analysis guides the existence proof in Section 4, where we first construct travelling wave solutions based on the asymptotic predictions, and then rigorously verify that these solutions indeed exhibit the asymptotic properties. Finally, in Section 5, we illustrate our results through three examples and corresponding numerical simulations.

2 Main results

Letting $u(x, t) = \phi(x - st) = \phi(\xi)$, it can be obtained from (1.16), (1.17) and (1.18) that $\phi(\xi)$ satisfies

$$\begin{cases} -s\phi_\xi - D\left(\frac{\phi_\xi}{\phi^{1-m}}\right)_\xi + f(\phi(\xi + sr))_\xi = 0, \\ \phi(\pm\infty) = u_\pm, \end{cases} \quad (2.1)$$

and

$$\begin{cases} -s\phi_\xi - D(\ln \phi)_{\xi\xi} + f(\phi(\xi + sr))_\xi = 0, \\ \phi(\pm\infty) = u_\pm, \end{cases} \quad (2.2)$$

and

$$\begin{cases} -s\phi_\xi - D\left(\frac{\phi_\xi}{\phi^{1+|m|}}\right)_\xi + f(\phi(\xi + sr))_\xi = 0, \\ \phi(\pm\infty) = u_\pm, \end{cases} \quad (2.3)$$

where s is the velocity of travelling waves $\phi(\xi)$.

Theorem 2.1 (Case $0 < m < 1$: Existence, monotonicity and oscillation of traveling waves)

For $0 < m < 1$, let

$$g(u) = f(u) - f(u_+) - s(u - u_+),$$

where the flux $f \in C^{k+1}(\mathbb{R})$ with $k \geq 0$. Suppose that $g'(u_+) = \cdots = g^{(k)}(u_+) = 0$ with $g^{(k+1)}(u_+) \neq 0$, and that s satisfies the Rankine-Hugoniot condition

$$s = \frac{f(0) - f(u_-)}{0 - u_-}, \quad (2.4)$$

and the entropy condition

$$f'(0) \leq s \leq f'(u_-). \quad (2.5)$$

1. **Existence:** the traveling wave solutions $\phi(x - st)$ connecting 0 and u_- exist for equation (2.1) under the following conditions:

- either $0 < s < f'(u_-)$, $0 < r < \bar{r}_1^{(1)}$, where

$$\bar{r}_1^{(1)} = \frac{D}{su_-^{1-m} \sqrt{f'(u_-)^2 - s^2}} \cdot \arctan\left(\frac{\sqrt{f'(u_-)^2 - s^2}}{s}\right);$$

- or $0 < s = f'(u_-)$ or $s < 0 = f'(u_-)$, for all time-delay $r > 0$.

2. **Asymptotic behaviors in the far fields:** let us define λ_1 as the smallest positive real root of

$$\lambda + \frac{su_-^{1-m}}{D} = \frac{f'(u_-)u_-^{1-m}}{D} e^{\lambda sr}, \quad (2.6)$$

and set

$$\Lambda_1^{(j)} = \min\{\operatorname{Re} \lambda > 0 \mid \lambda \in \mathbb{C}, \lambda \text{ satisfies (2.6), } (r, s) \in \Omega_1^{(j)}\}, \quad j = 1, 2,$$

where

$$\Omega_1^{(1)} = \{(r, s) : 0 < r < \bar{r}_1^{(1)}, 0 < s < f'(u_-)\},$$

$$\Omega_1^{(2)} = \{(r, s) : r \geq \hat{r}_1 = \frac{D}{sf'(u_-)u_-^{1-m}}, s = f'(u_-) > 0\}.$$

Then the asymptotic behaviors hold as $\xi \rightarrow \pm\infty$:

(i) ϕ approaches $u_+ = 0$ as $\xi \rightarrow +\infty$ in the following form:

- if $f'(0) \neq s$, then

$$|\phi| \sim \frac{1}{|\xi|^{\frac{1}{1-m}}}, \quad \text{as } \xi \rightarrow +\infty. \quad (2.7)$$

- if $f'(0) = s$, then

$$|\phi| \sim \frac{1}{|\xi|^{\frac{1}{k+1-m}}}, \quad \text{as } \xi \rightarrow +\infty. \quad (2.8)$$

(ii) ϕ approaches u_- exponentially as $\xi \rightarrow -\infty$.

3. **Monotonicity and Oscillation:** the criteria for monotonicity and oscillation of traveling waves are as follows:

- when $s < 0 = f'(u_-)$ for all $r > 0$, or $0 < s = f'(u_-)$ with $0 < r < \hat{r}_1$, then ϕ converges to u_- monotonically with exponential rate

$$u_- - \phi \sim e^{-\lambda_1|\xi|}, \quad \xi \rightarrow -\infty. \quad (2.9)$$

- when $0 < s < f'(u_-)$ for $0 < r < \bar{r}_1^{(1)}$, or $0 < s = f'(u_-)$ for $r \geq \hat{r}_1 = \frac{D}{sf'(u_-)u_-^{1-m}}$, then ϕ converges with decaying oscillations

$$|\phi - u_-| \sim e^{-\nu_1|\xi|}, \quad \xi \rightarrow -\infty, \quad (2.10)$$

where the decay rate $\nu_1 = \Lambda_1^{(1)}$ for $(r, s) \in \Omega_1^{(1)}$, or $\nu_1 = \Lambda_1^{(2)}$ for $(r, s) \in \Omega_1^{(2)}$.

Theorem 2.2 (Case $m = 0$: Existence, monotonicity and oscillation of traveling waves)

For $m = 0$, s, f and g satisfy the hypotheses of Theorem 2.1. We replace $1 - m$ by 1 in $\bar{r}_1^{(1)}$ and $\hat{r}_1$ to obtain $\bar{r}_2^{(1)}$ and $\hat{r}_2$, respectively.

1. **Existence:** the travelling wave solutions $\phi(x - st)$ connecting 0 and u_- exist for equation (2.2) under the following conditions:

- either $0 < s < f'(u_-)$ with $0 < r < \bar{r}_2^{(1)}$;
- or $0 < s = f'(u_-)$ or $s < 0 = f'(u_-)$, for all time-delay $r > 0$.

2. **Asymptotic behaviors in the far fields:** define λ_2 as the smallest positive real root of

$$\lambda + \frac{su_-}{D} = \frac{f'(u_-)u_-}{D} e^{\lambda sr}, \quad (2.11)$$

and

$$\Lambda_2^{(j)} = \min\{\operatorname{Re} \lambda > 0 \mid \lambda \in \mathbb{C}, \lambda \text{ satisfies (2.11)}, (r, s) \in \Omega_2^{(j)}\}, \quad j = 1, 2,$$

where

$$\Omega_2^{(1)} = \{(r, s) : 0 < r < \bar{r}_2^{(1)}, 0 < s < f'(u_-)\}, \quad \Omega_2^{(2)} = \{(r, s) : r \geq \hat{r}_2, s = f'(u_-) > 0\},$$

then the asymptotic decay properties as $\xi \rightarrow \pm\infty$ are:

- (i) the travelling wave solution converges to $u_+ = 0$ in the following forms:

- if $f'(0) \neq s$, then

$$|\phi| \sim \frac{1}{|\xi|}, \quad \text{as } \xi \rightarrow +\infty; \quad (2.12)$$

- if $f'(0) = s$, then

$$|\phi| \sim \frac{1}{|\xi|^{\frac{1}{k+1}}}, \quad \text{as } \xi \rightarrow +\infty; \quad (2.13)$$

(ii) the travelling wave solution converges to u_- exponentially as $\xi \rightarrow -\infty$.

3. Monotonicity and oscillation:

- when $s < 0 = f'(u_-)$ for all $r > 0$, or $0 < s = f'(u_-)$ with $0 < r < \hat{r}_2$, then ϕ converges to u_- monotonically in the form of

$$u_- - \phi \sim e^{-\lambda_2|\xi|}, \quad \xi \rightarrow -\infty; \quad (2.14)$$

- when $(r, s) \in \Omega_2^{(1)}$, or $(r, s) \in \Omega_2^{(2)}$, then ϕ converges to u_- with oscillations

$$|\phi - u_-| \sim e^{-\nu_2|\xi|}, \quad \xi \rightarrow -\infty, \quad (2.15)$$

where $\nu_2 = \Lambda_2^{(1)}$ for $(r, s) \in \Omega_2^{(1)}$, or $\nu_2 = \Lambda_2^{(2)}$ for $(r, s) \in \Omega_2^{(2)}$.

Theorem 2.3 (Case $m < 0$: Existence, monotonicity and oscillation of traveling waves)

For $m < 0$, s, f and g satisfy the hypotheses of Theorem 2.1. Let $\bar{r}_3^{(1)}$ and $\hat{r}_3$ be defined by replacing $1 - m$ with $1 + |m|$ in the expressions for $\bar{r}_1^{(1)}$ and $\hat{r}_1$, respectively.

1. **Existence:** the travelling wave solutions $\phi(x - st)$ connecting 0 and u_- exist for equation (2.3) under the following conditions:

- either $0 < s < f'(u_-)$ with $0 < r < \bar{r}_3^{(1)}$;
- or $0 < s = f'(u_-)$ or $s < 0 = f'(u_-)$, for all time-delay $r > 0$.

2. **Asymptotic behaviors in the far fields:** define λ_3 as the smallest positive real root of

$$\lambda + \frac{su_-^{1+|m|}}{D} = \frac{f'(u_-)u_-^{1+|m|}}{D} e^{\lambda sr}, \quad (2.16)$$

and

$$\Lambda_3^{(j)} = \min\{\operatorname{Re} \lambda > 0 \mid \lambda \in \mathbb{C}, \lambda \text{ satisfies (2.16)}, (r, s) \in \Omega_3^{(j)}\}, \quad j = 1, 2,$$

where

$$\Omega_3^{(1)} = \{(r, s) : 0 < r < \bar{r}_3^{(1)}, 0 < s < f'(u_-)\}, \quad \Omega_3^{(2)} = \{(r, s) : r \geq \hat{r}_3, s = f'(u_-) > 0\},$$

then

(i) the traveling wave $\phi(\xi)$ converges to $u_+ = 0$ as $\xi \rightarrow -\infty$ in the following forms:

- if $f'(0) \neq s$, then

$$|\phi| \sim \frac{1}{|\xi|^{\frac{1}{1+|m|}}}, \quad \text{as } \xi \rightarrow +\infty; \quad (2.17)$$

- if $f'(0) = s$, then

$$|\phi| \sim \frac{1}{|\xi|^{\frac{1}{k+1+|m|}}}, \quad \text{as } \xi \rightarrow +\infty; \quad (2.18)$$

(ii) the traveling wave $\phi(\xi)$ converges to u_- exponentially as $\xi \rightarrow -\infty$.

3. Monotonicity and oscillation:

- when $s < 0 = f'(u_-)$ for all $r > 0$ or $0 < s = f'(u_-)$ with $0 < r < \hat{r}_3$, then ϕ converges to u_- monotonically with

$$u_- - \phi \sim e^{-\lambda_3|\xi|}, \quad \xi \rightarrow -\infty; \quad (2.19)$$

- when $(r, s) \in \Omega_3^{(1)}$, or $(r, s) \in \Omega_3^{(2)}$, then ϕ converges to u_- with oscillations:

$$|\phi - u_-| \sim e^{-v_3|\xi|}, \quad \xi \rightarrow -\infty, \quad (2.20)$$

where $v_3 = \Lambda_3^{(1)}$ for $(r, s) \in \Omega_3^{(1)}$, or $v_3 = \Lambda_3^{(2)}$ for $(r, s) \in \Omega_3^{(2)}$.

3 Asymptotic analysis

This section performs formal asymptotic analysis to derive the expected decay rates of travelling waves as $\xi \rightarrow \pm\infty$.

We first analyze the behavior as $\xi \rightarrow +\infty$ through Taylor expansion, later we linearize equations for the cases $0 < m < 1$, $m = 0$, and $m < 0$ around $u = u_-$ to study the expected behavior as $\xi \rightarrow -\infty$. Utilizing lemmas on linear delay differential equations, this analysis yields conditions under which travelling waves are predicted to converge either monotonically or with oscillations, depending on parameters m , s , $f'(u_-)$ and r .

Lemma 3.1 For $0 < m < 1$, assume that s satisfies the Entropy condition (2.5), then the equation (2.6)

$$\lambda + \frac{su_-^{1-m}}{D} = \frac{f'(u_-)u_-^{1-m}}{D}e^{\lambda sr}$$

has one or two positive real roots if and only if s and $f'(u_-)$ satisfy the following conditions,

(i) $f'(u_-) = 0$ and $s < 0$.

(ii) $f'(u_-) \neq 0$,

- $s < f'(u_-) < 0$, or $s < 0 < f'(u_-)$.
- $s = f'(u_-) < 0$ with $r > \hat{r}_1$, or $0 < s = f'(u_-)$ with $0 < r < \hat{r}_1$, where

$$\hat{r}_1 = \frac{D}{sf'(u_-)u_-^{1-m}}.$$

- $0 < s < f'(u_-)$ with $0 < r \leq r_1^*$, where r_1^* is the smaller solution of

$$re^{-s^2ru_-^{1-m}/D} = \frac{D}{sf'(u_-)u_-^{1-m}e}. \quad (3.1)$$

Proof. If $f'(u_-) = 0$, we can get that

$$\lambda = -\frac{su_-^{1-m}}{D},$$

thus (2.6) has a positive root λ when $s < 0 = f'(u_-)$.

When $f'(u_-) \neq 0$, if $s < f'(u_-) < 0$, or $s < 0 < f'(u_-)$, it is easy to find that the equation (2.6) has a positive root.

Suppose that the two functions with respect to λ on both sides of the equation (2.6) are tangent at some point (λ, s) , we have

$$\begin{cases} \lambda + \frac{su_-^{1-m}}{D} = \frac{f'(u_-)u_-^{1-m}}{D} e^{\lambda sr}, \\ 1 = \frac{sr f'(u_-)u_-^{1-m}}{D} e^{\lambda sr}. \end{cases} \quad (3.2)$$

If $s = f'(u_-) < 0$, then $\lambda = 0$ is a root of (2.6), the two functions are tangent at the point $(0, s)$, then we obtain $\frac{sr f'(u_-)u_-^{1-m}}{D} = 1$ from the second equation of (3.2) and denote

$$\hat{r}_1 = \frac{D}{s f'(u_-)u_-^{1-m}}.$$

Through a simple analysis to two functions, we obtain that if $r > \hat{r}_1$, then (2.6) has two roots $\lambda_a = 0, \lambda_b > 0$; if $r < \hat{r}_1$, then (2.6) has two roots $\lambda_a = 0, \lambda_b < 0$. The situation is similar for $0 < s = f'(u_-)$, if $r > \hat{r}_1$, then (2.6) has two roots $\lambda_a = 0, \lambda_b < 0$; if $r < \hat{r}_1$, then (2.6) has two roots $\lambda_a = 0, \lambda_b > 0$.

If $0 < s < f'(u_-)$, in view of (3.2), we can calculate that

$$\lambda = \frac{1}{sr} \ln \frac{D}{sr f'(u_-)u_-^{1-m}}. \quad (3.3)$$

In order to get the positive root of the equation (2.6), r must satisfies $r < \hat{r}_1$ because of $s > 0$ and (3.3). Substitute (3.3) into the first equation of (3.2), we see

$$\frac{D}{sru_-^{1-m}} \ln \frac{D}{sr f'(u_-)u_-^{1-m}} + s = \frac{D}{sru_-^{1-m}},$$

simplifying it yields the equation with respect to r

$$e^{-s^2 ru_-^{1-m}/D} = \frac{D}{sr f'(u_-)u_-^{1-m} e}, \quad (3.4)$$

thus, the equation in terms of λ is transformed into an equation in terms of r . Since $r < \hat{r}_1$, we can see that

$$\frac{D}{sr f'(u_-)u_-^{1-m} e} > \frac{1}{e}.$$

When $0 < r < D/s^2 u_-^{1-m}$,

$$e^{-s^2 ru_-^{1-m}/D} > \frac{1}{e},$$

which implies that the functions on both sides of the equation (3.1) will intersect at some points.

Define $g(r) \triangleq r e^{-s^2 ru_-^{1-m}/D}$, $r \geq 0$, the solutions of (3.4) are equivalent to the solutions of $g(r) = D/sf'(u_-)u_-^{1-m}e$, which is exactly equation (3.1), we derive that

$$g'(r) = e^{-s^2 ru_-^{1-m}/D} \left(1 - \frac{s^2 ru_-^{1-m}}{D} \right).$$

Setting $g'(r) = 0$ yields the unique critical point satisfy

$$r_{\max} = \frac{D}{s^2 u_-^{1-m}} > \frac{D}{s f'(u_-)u_-^{1-m}} = \hat{r}_1.$$

Clearly $g'(r) > 0$ for $r < r_{\max}$ and $g'(r) < 0$ for $r > r_{\max}$, hence $g(r)$ attains its maximum at $r = r_{\max}$. The corresponding maximum value satisfies

$$g(r_{\max}) = \frac{D}{s^2 u_-^{1-m} e} > \frac{D}{s f'(u_-) u_-^{1-m} e},$$

where the inequality follows from the assumption $0 < s < f'(u_-)$. Consequently, equation (3.1) admits two positive solutions, one on each side of $r_{\max}$. Since $\hat{r}_1 < r_{\max}$, the larger root lies outside the feasible range, and the smaller root r_1^* of (3.1) must lie in the interval $(0, \hat{r}_1)$ for the following reason.

Evaluating $g(r)$ at $r = \hat{r}_1$ gives

$$g(\hat{r}_1) = \frac{D}{s f'(u_-) u_-^{1-m}} e^{-\frac{s}{f'(u_-)}} > \frac{D}{s f'(u_-) u_-^{1-m} e} = g(r_1^*),$$

for $0 < s < f'(u_-)$. Since $g(r)$ is strictly increasing on $(0, r_{\max})$, the inequality $g(\hat{r}_1) > g(r_1^*)$ implies $r_1^* < \hat{r}_1$.

By analyzing the two function graphs on both sides of the equation (2.6), if $r < r_1^*$, then (2.6) has two positive roots; if $r = r_1^*$, then (2.6) has exactly one positive root; if $r > r_1^*$, no solution exists. Thus, the proof of Lemma 3.1 is complete. $\square$

Lemma 3.2 For $m = 0$, suppose that s satisfies $f'(0) \leq s \leq f'(u_-)$, then equation (2.11)

$$\lambda + \frac{su_-}{D} = \frac{f'(u_-)u_-}{D} e^{\lambda sr}$$

has one or two positive real roots if and only if s and $f'(u_-)$ satisfy the following conditions,

- (i) $f'(u_-) = 0$ and $s < 0$.
- (ii) $f'(u_-) \neq 0$,
 - $s < f'(u_-) < 0$, or $s < 0 < f'(u_-)$.
 - $s = f'(u_-) < 0$ with $r > \hat{r}_2$, or $0 < s = f'(u_-)$ with $0 < r < \hat{r}_2$, where

$$\hat{r}_2 = \frac{D}{s f'(u_-) u_-}.$$

- $0 < s < f'(u_-)$ with $0 < r \leq r_2^*$, where r_2^* is the smaller solution of

$$r e^{-s^2 r u_- / D} = \frac{D}{s f'(u_-) u_- e}. \quad (3.5)$$

Lemma 3.3 Given that $m < 0$, and s satisfies $f'(0) \leq s \leq f'(u_-)$, then equation (2.16)

$$\lambda + \frac{su_-^{1+|m|}}{D} = \frac{f'(u_-)u_-^{1+|m|}}{D} e^{\lambda sr}$$

has one or two positive roots if and only if s and $f'(u_-)$ satisfy the following conditions,

- (i) $f'(u_-) = 0$ and $s < 0$.
- (ii) $f'(u_-) \neq 0$,
 - $s < f'(u_-) < 0$, or $s < 0 < f'(u_-)$.

- $s = f'(u_-) < 0$ with $r > \hat{r}_3$, or $0 < s = f'(u_-)$ with $0 < r < \hat{r}_3$, where $\hat{r}_3 = \frac{D}{sf'(u_-)u_-^{1+|m|}}$.
- $0 < s < f'(u_-)$ with $0 < r \leq r_3^*$, where r_3^* is the smaller solution of

$$re^{-s^2ru_-^{1+|m|}/D} = \frac{D}{sf'(u_-)u_-^{1+|m|}e}. \quad (3.6)$$

The proofs of Lemmas 3.2 and 3.3 closely follow that of Lemma 3.1, thus the details are omitted.

The scalar linear delayed differential equation is

$$\begin{cases} z'(t) + k_1 z(t) = k_2 z(t-r), \\ z(\tau) = z_0(\tau), \quad \tau \in [-r, 0], \end{cases} \quad (3.7)$$

where k_1, k_2 , and r are constants with $r > 0$. The characteristic equation of (3.7) is

$$\lambda + k_1 - k_2 e^{-\lambda r} = 0. \quad (3.8)$$

The scalar equation (3.7) has a nontrivial solution $Ce^{\lambda t}$ ($C \in \mathbb{R}$) if and only if λ satisfies the characteristic equation (3.8). When λ is a real root, no oscillation occurs, we have discussed this in Lemmas 3.1, 3.2 and 3.3. When λ is a complex root, oscillation may arise. Therefore, we focus on when the equation (3.8) possesses complex roots.

Lemma 3.4 [see [11]] *If there is a sequence $\{\lambda_j\}$ of solutions of equation (3.8) such that $|\lambda_j| \rightarrow \infty$ as $j \rightarrow \infty$, then $\operatorname{Re} \lambda_j \rightarrow -\infty$ as $j \rightarrow \infty$. Thus, there is a real number α such that all solutions of equation (3.8) satisfy $\operatorname{Re} \lambda < \alpha$ and there are only a finite number of solutions in any vertical strip in the complex plane.*

Lemma 3.5 *Consider the characteristic equation (3.8), as $r \rightarrow 0^+$, then there exists exactly one root $\lambda_1(r)$ such that*

$$\lim_{r \rightarrow 0^+} \lambda_1(r) = k_2 - k_1,$$

while all other roots $\lambda(r)$ satisfy

$$\lim_{r \rightarrow 0^+} |\lambda(r)| = \infty.$$

Proof. When $r = 0$, the equation reduces to $\lambda + k_1 = k_2$, which yields the unique root $\lambda_0 = k_2 - k_1$.

For $r > 0$, define

$$F(\lambda, r) = \lambda + k_1 - k_2 e^{-\lambda r} = 0. \quad (3.9)$$

At $(\lambda_0, 0)$, we have $F(\lambda_0, 0) = 0$ and $\frac{\partial F}{\partial \lambda}(\lambda_0, 0) = 1 \neq 0$. By the implicit function theorem, there exists a unique continuous function $\lambda_1(r)$ for small $r \geq 0$ such that $F(\lambda_1(r), r) = 0$ and $\lambda_1(0) = \lambda_0$.

Now suppose, for contradiction, that some root $\lambda(r)$ remains bounded as $r \rightarrow 0^+$. Then along some sequence $r_n \rightarrow 0^+$, we have $\lambda(r_n) \rightarrow \tilde{\lambda}$ for some finite $\tilde{\lambda}$. Taking the limit in the characteristic equation gives

$$\tilde{\lambda} + k_1 = k_2 e^{-\tilde{\lambda} \cdot 0} = k_2,$$

so $\tilde{\lambda} = k_2 - k_1 = \lambda_0$. Hence both $\lambda_1(r)$ and $\lambda(r)$ converge to λ_0 as $r \rightarrow 0^+$. However, when $r = 0$, the equation is linear and has only the simple root λ_0 , since $\frac{\partial F}{\partial \lambda}(\lambda_0, 0) = 1 \neq 0$. This contradiction implies that all other roots must satisfy $|\lambda(r)| \rightarrow \infty$ as $r \rightarrow 0^+$. $\square$

The following lemma addresses the case $k_1 < 0$, which corresponds to the regime $s > 0$ in our travelling wave analysis.

Lemma 3.6 For linear delayed differential equation (3.7), $k_1 < 0$,

- (i) For $k_2 < 0$, $|k_2| > |k_1|$, the real parts of complex roots of characteristic equation (3.8) are negative if and only if $0 < r < \bar{r}_1$, where

$$\bar{r}_1 = \left[\arctan\left(|k_1|^{-1} \sqrt{k_2^2 - k_1^2}\right) \right] / \sqrt{k_2^2 - k_1^2}.$$

- (ii) For $k_2 > 0$, $|k_2| > |k_1|$, the real parts of complex roots of characteristic equation (3.8) are negative if and only if $0 < r < \bar{r}_2$, where

$$\bar{r}_2 = \left[\arctan\left(|k_1|^{-1} \sqrt{k_2^2 - k_1^2}\right) + \pi \right] / \sqrt{k_2^2 - k_1^2}.$$

- (iii) For $|k_2| \leq |k_1|$, it holds that the real parts of all complex roots of equation (3.8) are negative.

Proof. Assume that $\lambda = \alpha + i\beta$ is a complex root of (3.8), then one of the solutions of (3.7) is

$$e^{\alpha t} [C_1 \cos(\beta t) + C_2 \sin(\beta t)]. \quad (3.10)$$

When $\alpha < 0$, the solution (3.10) decays exponentially and will grow exponentially if $\alpha > 0$. We need to determine under what conditions the real part of λ is positive or negative. Suppose that $\lambda = i\omega$, $\omega > 0$, and substitute it into (3.8), we can derive that

$$\begin{cases} k_1 = k_2 \cos(\omega r), \\ \omega = -k_2 \sin(\omega r). \end{cases} \quad (3.11)$$

From (3.11) it follows that $k_1^2 + \omega^2 = k_2^2$, $\omega = \sqrt{k_2^2 - k_1^2}$ and

$$\tan(\omega r) = \frac{\sin(\omega r)}{\cos(\omega r)} = \frac{-\omega/k_2}{k_1/k_2} = -\frac{\omega}{k_1} > 0,$$

then we obtain

$$\omega r = \arctan\left(\frac{\omega}{|k_1|}\right) + N\pi, \quad N \in \mathbb{Z}.$$

Since $|\cos(\omega r)| = |k_1/k_2| \leq 1$, thus $|k_1| \leq |k_2|$. If $|k_1| = |k_2|$, then $\omega = 0$ contradicts the requirement that $\omega > 0$, which implies that purely imaginary root exists if and only if $|k_1| < |k_2|$.

If $k_2 < 0$, thus $\cos(\omega r) = k_1/k_2 > 0$, $\sin(\omega r) = -\omega/k_2 > 0$, here ωr is located in the first quadrant, take $N = 2n$,

$$\bar{r}_{1,n} = \left[\arctan\left(\frac{\omega}{|k_1|}\right) + 2n\pi \right] / \omega, \quad n \in \mathbb{Z}.$$

If $k_2 > 0$, $\cos(\omega r) = k_1/k_2 < 0$, $\sin(\omega r) = -\omega/k_2 < 0$, thus ωr is located in the third quadrant, take $N = 2n + 1$,

$$\bar{r}_{2,n} = \left[\arctan\left(\frac{\omega}{|k_1|}\right) + (2n + 1)\pi \right] / \omega, \quad n \in \mathbb{Z},$$

then we obtain $\bar{r}_1$ and $\bar{r}_2$ respectively when $n = 0$. In view of (3.9), we can see that

$$\frac{d\lambda}{dr} = -\frac{\partial F / \partial r}{\partial F / \partial \lambda} = -\frac{k_2 \lambda e^{-\lambda r}}{1 + k_2 r e^{-\lambda r}}.$$

Substitute $\lambda = i\omega$ into the above equality, it is easy to calculate that

$$\operatorname{Re} \left(\frac{d\lambda}{dr} \right) \Big|_{\lambda=i\omega} = \frac{\omega^2}{(1 + rk_1)^2 + (\omega r)^2} > 0.$$

For $k_2 < 0$, if purely imaginary roots exist, then the real part of λ is negative when $0 < r < \bar{r}_1$, while the corresponding complex roots have positive real parts when $r > \bar{r}_1$. The conclusion is similar when $k_2 > 0$.

Next, we consider the case where no purely imaginary roots exist, and it follows from the preceding derivation that $|k_1| \geq |k_2|$. Employing Lemma 3.4 and 3.5, we conclude that

$$\operatorname{Re} \lambda \rightarrow -\infty, \quad \text{as } r \rightarrow 0^+.$$

These complex roots vary continuously with r . Suppose that, for some $r_1 > 0$, the real part a of a certain complex root turns positive $a > 0$. Since a depends continuously on r , there exists some $r_0 > 0$ such that $a = 0$ due to the Intermediate Value Theorem, implying the root becomes purely imaginary. This yields a contradiction. Consequently, the real part a of every complex root must remain strictly negative for all $r > 0$. Thus, the proof of Lemma 3.6 is complete. $\square$

Before proving the main theorems, we perform a linearized analysis to determine the expected asymptotic behavior of travelling waves.

Proposition 3.1 *For $0 < m < 1$, under the hypotheses of Theorem 2.1, we perform asymptotic analysis of equation (2.1) near the constant states 0 and u_- , then the properties of ϕ around 0 are*

- $f'(0) \neq s$,

$$|\phi| \sim \frac{1}{|\xi|^{\frac{1}{1-m}}}, \quad \text{as } \xi \rightarrow +\infty. \quad (3.12)$$

- $f'(0) = s$,

$$|\phi| \sim \frac{1}{|\xi|^{\frac{1}{k+1-m}}}, \quad \text{as } \xi \rightarrow +\infty. \quad (3.13)$$

The exponentially convergent profile near u_- is classified as follows,

(i) *monotone*

- $s < 0 = f'(u_-)$, for all time-delay $r > 0$.
- $0 < s = f'(u_-)$, $0 < r < \hat{r}_1$.

(ii) *oscillatory*

- $0 < s < f'(u_-)$, $0 < r < \bar{r}_1^{(1)}$.
- $0 < s = f'(u_-)$, $r \geq \hat{r}_1$.

The exact exponential decay rates in each case are given by the characteristic equation (2.6) and are denoted as in Theorem 2.1.

Proposition 3.2 *For $m = 0$, under the hypotheses of Theorem 2.2, we perform asymptotic analysis of equation (2.2) near the constant states 0 and u_- , the behavior of ϕ around 0 is*

- $f'(0) \neq s$,

$$|\phi| \sim \frac{1}{|\xi|}, \quad \text{as } \xi \rightarrow +\infty. \quad (3.14)$$

- $f'(0) = s$,

$$|\phi| \sim \frac{1}{|\xi|^{\frac{1}{k+1}}}, \quad \text{as } \xi \rightarrow +\infty. \quad (3.15)$$

The exponential approach to u_- is further refined into either monotone or oscillatory character as follows,

(i) *monotone*

- $s < 0 = f'(u_-)$, for all time-delay $r > 0$.
- $0 < s = f'(u_-)$, $0 < r < \hat{r}_2$.

(ii) *oscillatory*

- $0 < s < f'(u_-)$, $0 < r < \bar{r}_2^{(1)}$.
- $0 < s = f'(u_-)$, $r \geq \hat{r}_2$.

The rates follow from the characteristic equation (2.11) and are given explicitly in Theorem 2.2.

Proposition 3.3 For $m < 0$, under the hypotheses of Theorem 2.3, we perform asymptotic analysis of equation (2.3) near the constant states 0 and u_- , the profile of ϕ surrounding 0 is

- $f'(0) \neq s$,

$$|\phi| \sim \frac{1}{|\xi|^{\frac{1}{1+m}}}, \quad \text{as } \xi \rightarrow +\infty. \quad (3.16)$$

- $f'(0) = s$,

$$|\phi| \sim \frac{1}{|\xi|^{\frac{1}{k+1+m}}}, \quad \text{as } \xi \rightarrow +\infty. \quad (3.17)$$

ϕ converges to u_- exponentially, splitting into monotone and oscillatory cases,

(i) *monotone*

- $s < 0 = f'(u_-)$, for all time-delay $r > 0$.
- $0 < s = f'(u_-)$, $0 < r < \hat{r}_3$.

(ii) *oscillatory*

- $0 < s < f'(u_-)$, $0 < r < \bar{r}_3^{(1)}$.
- $0 < s = f'(u_-)$, $r \geq \hat{r}_3$.

The decay rates are as established in Theorem 2.3.

Lemma 3.7 Assume that $0 < s < f'(u_-)$, then we can obtain that

$$\hat{r}_1 < \bar{r}_1^{(1)}.$$

Moreover, the analogous inequalities

$$\hat{r}_2 < \bar{r}_2^{(1)}, \quad \hat{r}_3 < \bar{r}_3^{(1)}$$

also hold.

Proof. According to the expressions of $\bar{r}_1^{(1)}$ and $\hat{r}_1$, we can see that

$$\frac{\bar{r}_1^{(1)}}{\hat{r}_1} = \frac{\arctan\left(\frac{\sqrt{f'(u_-)^2 - s^2}}{s}\right)}{\sqrt{f'(u_-)^2 - s^2}} \cdot f'(u_-),$$

denote $x = \sqrt{f'(u_-)^2 - s^2}/s > 0$, then

$$\frac{\bar{r}_1^{(1)}}{\hat{r}_1} = s \sqrt{1+x^2} \frac{\arctan x}{sx} = \frac{\sqrt{1+x^2}}{x} \arctan x.$$

Assume that $g(x) = \frac{\sqrt{1+x^2}}{x} \arctan x$, $x > 0$,

$$g'(x) = \frac{x - \arctan x}{x^2 \sqrt{1+x^2}} > 0, \quad \forall x > 0.$$

Since $g'(x) > 0$ and $g(0^+) = 1$, it follows that $g(x) > 1$ for all $x > 0$, which implies that $\hat{r}_1 < \bar{r}_1^{(1)}$. Similarly, we can obtain $\hat{r}_2 < \bar{r}_2^{(1)}$ and $\hat{r}_3 < \bar{r}_3^{(1)}$. The proof of Lemma 3.7 is complete. $\square$

Suppose that equation (3.8) has a real root λ and complex roots $\alpha \pm i\beta$, then the solution of (3.7) is

$$z(t) = C_1 e^{\lambda t} + e^{\alpha t} [C_2 \cos(\beta t) + C_3 \sin(\beta t)].$$

If $\lambda, \alpha < 0$, whether $z(t)$ is monotonic or oscillatory as $t \rightarrow +\infty$ is determined by which of them has the smaller absolute value. The asymptotic behavior of the traveling wave solution as $\xi \rightarrow -\infty$ is then investigated by applying the lemmas previously stated, and the analysis established above.

Proof of Proposition 3.1. For $0 < m < 1$, formally we have

$$\lim_{\xi \rightarrow \pm\infty} \frac{\phi_\xi}{\phi^{1-m}} = 0,$$

integrating the first equation of (2.1) from $-\infty$ to $+\infty$, then we obtain

$$-s\phi \Big|_{-\infty}^{+\infty} - D \frac{\phi_\xi}{\phi^{1-m}} \Big|_{-\infty}^{+\infty} + f(\phi(\xi + sr)) \Big|_{-\infty}^{+\infty} = 0.$$

It can be calculated from the above equation that

$$-s(0 - u_-) + f(0) - f(u_-) = 0,$$

then

$$s = \frac{f(0) - f(u_-)}{0 - u_-},$$

which implies that s satisfies the Rankine-Hugoniot condition (2.4). Integrating (2.1) from $-\infty$ to ξ and from ξ to $+\infty$, then we calculate that

$$-s(\phi - u_\pm) - D \frac{\phi_\xi}{\phi^{1-m}} + f(\phi(\xi + sr)) - f(u_\pm) = 0, \quad (3.18)$$

which implies that

$$\phi_\xi = \frac{1}{D} \phi^{1-m} [f(\phi(\xi + sr)) - f(u_\pm) - s(\phi - u_\pm)]. \quad (3.19)$$

Since oscillation could occur around u_- , we first investigate the asymptotic behavior of ϕ as $\xi \rightarrow +\infty$.

When $f'(0) \neq s$,

$$\begin{aligned}\phi_\xi &= \frac{1}{D} \phi^{1-m} [f(\phi(\xi + sr)) - f(0) - s(\phi - 0)] \\ &= \frac{1}{D} \phi^{2-m} \left[\frac{f(\phi(\xi + sr)) - f(0)}{\phi - 0} - s \right].\end{aligned}\quad (3.20)$$

Assume that $\phi(\xi) = \alpha \xi^{-\beta}$, $\beta > 0$, thus $\lim_{\xi \rightarrow +\infty} \frac{\phi(\xi + sr)}{\phi(\xi)} = 1$, and

$$\lim_{\xi \rightarrow +\infty} \frac{f(\phi(\xi + sr)) - f(0)}{\phi - 0} = \lim_{\xi \rightarrow +\infty} \frac{f(\phi(\xi + sr)) - f(0)}{\phi(\xi + sr) - 0} \frac{\phi(\xi + sr)}{\phi(\xi)} = f'(0).$$

Letting $\phi \rightarrow 0^+$ ($\xi \rightarrow +\infty$) in (3.20), we deduced that $\phi_\xi \sim (f'(0) - s) \phi^{2-m}$ as $\xi \rightarrow +\infty$, thus

$$\frac{d\phi}{\phi^{2-m}} = O(1) d\xi.$$

Then we can calculate that $\phi = O(1) \frac{1}{|\xi|^{\frac{1}{1-m}}}$, as $\xi \rightarrow +\infty$, which implies that $\beta = \frac{1}{1-m}$.

If $f'(0) = s$, then by the definition of $g(u)$ and the assumption on $g(u)$ in Theorem 2.1, we have $f''(0) = \dots = f^{(k)}(0) = 0$ with $f^{(k+1)}(0) \neq 0$, we deduce that

$$\begin{aligned}\phi_\xi &= \frac{1}{D} \phi^{1-m} [f(\phi(\xi + sr)) - f(0) - s(\phi - 0)] \\ &= \frac{1}{D} \phi^{1-m} \left[f'(0)(\phi(\xi + sr) - \phi(\xi)) + \frac{f^{(k+1)}(0)}{(k+1)!} \phi^{k+1}(\xi) \frac{\phi^{k+1}(\xi + sr)}{\phi^{k+1}(\xi)} + o(\phi^{k+2}(\xi + sr)) \right]\end{aligned}$$

by utilizing Taylor's expansion. We also assume that $\phi(\xi) = \alpha \xi^{-\beta}$, $\beta > 0$, for the above expression is an algebraic expression and $\lim_{\xi \rightarrow +\infty} \frac{\phi(\xi + sr)}{\phi(\xi)} = 1$, we can derive that $|\phi_\xi| \sim |\phi|^{k+2-m}$, which implies that

$$\phi = O(1) \frac{1}{|\xi|^{\frac{1}{k+1-m}}}, \quad \text{as } \xi \rightarrow +\infty, \quad f'(0) = s.$$

Next we investigate the asymptotic behavior and oscillatory of ϕ around $u = u_-$. If $s > 0$, linearizing equation (3.19) around u_- ,

$$\phi_\xi + \frac{su_-^{1-m}}{D} (\phi - u_-) = \frac{f'(u_-)u_-^{1-m}}{D} (\phi(\xi + sr) - u_-). \quad (3.21)$$

Then the characteristic equation of (3.21) is equation (2.6)

$$\lambda + \frac{su_-^{1-m}}{D} = \frac{f'(u_-)u_-^{1-m}}{D} e^{\lambda sr},$$

the existence of positive real roots λ of the above equation is obtained by Lemma 3.1. If $f'(u_-) = 0$, equation (3.21) reduces to

$$\phi_\xi + \frac{su_-^{1-m}}{D} (\phi - u_-) = 0,$$

thus we can obtain that

$$u_- - \phi \leq C e^{-\frac{su_-^{1-m}}{D}\xi}, \quad \text{as } \xi \rightarrow -\infty.$$

Hence ϕ converge exponentially to u_- if and only if $s < 0$, here $-\frac{su_-^{1-m}}{D}$ is a root of the above characteristic equation.

If $f'(u_-) \neq 0$, we first investigate $s > 0$, denote $\phi(\xi) = y(\xi) + u_-$, we see

$$y_\xi + \frac{su_-^{1-m}}{D}y = \frac{f'(u_-)u_-^{1-m}}{D}y(\xi + sr).$$

Letting $z(t) = y(\xi(t))$, $\xi(t) = -t$, then $y(\xi + sr) = y(-t + sr) = z(t - sr)$ and we obtain

$$\frac{dz}{dt} = \frac{dy}{d\xi} \frac{d\xi}{dt} = -y_\xi,$$

hence

$$z_t - \frac{su_-^{1-m}}{D}z = -\frac{f'(u_-)u_-^{1-m}}{D}z(t - sr), \quad (3.22)$$

then the characteristic equation of (3.22) is

$$\lambda - \frac{su_-^{1-m}}{D} = -\frac{f'(u_-)u_-^{1-m}}{D}e^{-\lambda sr}. \quad (3.23)$$

Define

$$k_1^{(1)} = -\frac{su_-^{1-m}}{D} < 0, \quad k_2^{(1)} = -\frac{f'(u_-)u_-^{1-m}}{D},$$

we look for roots of equation (3.23) with negative real parts because of the change of variable $\xi(t) = -t$.

Since $0 < s < f'(u_-)$, then $k_1^{(1)}, k_2^{(1)} < 0$ and $|k_2^{(1)}| > |k_1^{(1)}|$. Employing Lemma 3.6, we can obtain that if $0 < r < \bar{r}_1^{(1)}$, the real parts of complex roots λ of (3.23) are negative, where

$$\begin{aligned} \bar{r}_1^{(1)} &:= \left[\arctan \left(|k_1^{(1)}|^{-1} \sqrt{|k_2^{(1)}|^2 - |k_1^{(1)}|^2} \right) \right] / \left(s \sqrt{|k_2^{(1)}|^2 - |k_1^{(1)}|^2} \right) \\ &= \frac{D}{su_-^{1-m} \sqrt{f'(u_-)^2 - s^2}} \cdot \arctan \left(\frac{\sqrt{f'(u_-)^2 - s^2}}{s} \right). \end{aligned}$$

Thanks to the transformation $\xi(t) = -t$, the complex roots λ of (2.6) have positive real parts when $0 < r < \bar{r}_1^{(1)}$. Assume $\lambda = p_1 + iq_1$ with $p_1 > 0$, $q_1 > 0$, and substitute it into (2.6). Separating real and imaginary parts yields the equations

$$\begin{cases} p_1 + \frac{su_-^{1-m}}{D} = \frac{f'(u_-)u_-^{1-m}}{D} \cos(q_1 sr) e^{p_1 sr}, \\ q_1 = \frac{f'(u_-)u_-^{1-m}}{D} \sin(q_1 sr) e^{p_1 sr}. \end{cases}$$

For $p_1 > 0$ and $0 < s < f'(u_-)$, we have $\cos(q_1 sr) > 0$, moreover $\cos(q_1 sr) < 1$ since $q_1 > 0$ and $\sin(q_1 sr) \neq 0$.

Recalling Lemma 3.1, for $0 < s < f'(u_-)$, equation (2.6) possesses exactly one positive real root when $r < r_1^*$ and two positive real roots when $r = r_1^*$. Let λ_1 denote the smallest among these positive real roots (in either case). When both real and complex characteristic

roots coexist, whether ϕ converges monotonically or oscillates toward u_- as $\xi \rightarrow -\infty$ is determined by the smaller of λ_1 and p_1 .

To compare λ_1 and p_1 , introduce the auxiliary functions

$$H(x) = x + \frac{su_-^{1-m}}{D}, \quad Q_1(x) = \frac{f'(u_-)u_-^{1-m}}{D}e^{xsr}, \quad Q_2(x) = \frac{f'(u_-)u_-^{1-m}}{D}\cos(q_1sr)e^{xsr}.$$

The real root λ_1 satisfies $H(\lambda_1) = Q_1(\lambda_1)$, whereas the real part p_1 of a complex root satisfies $H(p_1) = Q_2(p_1)$. Since $0 < \cos(q_1sr) < 1$, the curve $Q_2(x)$ lies strictly below $Q_1(x)$ for all x . Consequently, the intersection of $H(x)$ with $Q_2(x)$ occurs at a smaller x than its intersection with $Q_1(x)$. Hence $\lambda_1 > p_1 > 0$.

From Lemmas 3.1 and 3.7 we obtain $r_1^* < \hat{r}_1 < \bar{r}_1^{(1)}$. Since $p_1 < \lambda_1$, the solution ϕ approaches u_- with oscillations when $0 < r < \bar{r}_1^{(1)}$, and its decay rate is governed by the complex root with the smallest positive real part. Thus

$$|\phi - u_-| \leq Ce^{\Lambda_1^{(1)}\xi}, \quad \xi \rightarrow -\infty,$$

where the decay rate $\Lambda_1^{(1)}$ is defined by

$$\Lambda_1^{(1)} = \min \left\{ \operatorname{Re} \lambda \mid \lambda \in \mathbb{C}, \operatorname{Re} \lambda > 0, \lambda + \frac{su_-^{1-m}}{D} = \frac{f'(u_-)u_-^{1-m}}{D}e^{-\lambda sr}, (r, s) \in \Omega_1^{(1)} \right\},$$

with the parameter set

$$\Omega_1^{(1)} = \{(r, s) \mid 0 < r < \bar{r}_1^{(1)}, 0 < s < f'(u_-)\}.$$

If $s = f'(u_-) > 0$, Lemma 3.1 shows that equation (2.6) has a zero root and a positive real root λ_1 for $0 < r < \hat{r}_1$. In this case $|k_1^{(1)}| = |k_2^{(1)}|$, and by Lemma 3.6 all complex roots of (3.23) have negative real parts for any $r > 0$. Because $\xi(t) = -t$, every complex root of (2.6) therefore possesses a positive real part. Assume there are two complex roots, denoted by $p_1 + iq_1$ with $p_1, q_1 > 0$ and $p_2 + iq_2$ with $p_2 < 0$ and $q_2 > 0$, which satisfy equation (2.6). Then $H(0) = Q_1(0)$, $H(\lambda_1) = Q_1(\lambda_1)$, $H(p_1) = Q_2(p_1)$, and $H(p_2) = Q_2(p_2)$. Again $Q_2(x)$ lies below $Q_1(x)$, and the analysis of the curves shows that the x -coordinates of the intersections satisfy $0 < \lambda_1 < p_1$ and $p_2 < 0$.

When $0 < s = f'(u_-)$ and $0 < r < \hat{r}_1$, the decay comes from the positive real root λ_1 of characteristic equation (2.6), giving

$$u_- - \phi \leq Ce^{\lambda_1\xi}, \quad \xi \rightarrow -\infty.$$

For $r \geq \hat{r}_1$, equation (2.6) possesses no real roots, leading to oscillations, then

$$|\phi - u_-| \leq Ce^{\Lambda_1^{(2)}\xi}, \quad \xi \rightarrow -\infty,$$

where

$$\Lambda_1^{(2)} = \min \left\{ \operatorname{Re} \lambda \mid \lambda \in \mathbb{C}, \operatorname{Re} \lambda > 0, \lambda + \frac{su_-^{1-m}}{D} = \frac{f'(u_-)u_-^{1-m}}{D}e^{-\lambda sr}, (r, s) \in \Omega_1^{(2)} \right\},$$

with

$$\Omega_1^{(2)} = \{(r, s) \mid r \geq \hat{r}_1, 0 < s = f'(u_-)\}.$$

If $s < 0$, we review (3.21) and obtain

$$\frac{D}{\phi^{1-m}} \phi_\xi - |s|(\phi - u_-) = f(\phi(\xi - |s|r)) - f(u_-),$$

Linearizing the above equation around u_- , we can obtain that

$$\phi_\xi - \frac{|s|u_-^{1-m}}{D}(\phi - u_-) = \frac{f'(u_-)u_-^{1-m}}{D}(\phi(\xi - |s|r) - u_-).$$

the characteristic equation is equation (2.6)

$$\lambda - \frac{|s|u_-^{1-m}}{D} = \frac{f'(u_-)u_-^{1-m}}{D} e^{-\lambda|s|r}.$$

Denote

$$k_1^{(2)} = -\frac{|s|u_-^{1-m}}{D} < 0, \quad k_2^{(2)} = \frac{f'(u_-)u_-^{1-m}}{D}.$$

For $f'(u_-) > -s > 0 > s$, then $k_2^{(2)} > 0$ and $|k_2^{(2)}| > |k_1^{(2)}|$. Utilizing Lemma 4.3, if $0 < r < \bar{r}_1^{(2)}$, the real parts of complex roots of equation (2.6) are negative, and at least one complex root of (2.6) has a positive real part if $r > \bar{r}_1^{(2)}$, where

$$\begin{aligned} \bar{r}_1^{(2)} &:= \left[\arctan \left(|k_1^{(2)}|^{-1} \sqrt{|k_2^{(2)}|^2 - |k_1^{(2)}|^2} \right) + \pi \right] / |s| \sqrt{|k_2^{(2)}|^2 - |k_1^{(2)}|^2} \\ &= \frac{D}{|s|u_-^{1-m} \sqrt{f'(u_-)^2 - s^2}} \cdot \left[\arctan \left(\frac{\sqrt{f'(u_-)^2 - s^2}}{|s|} \right) + \pi \right]. \end{aligned}$$

Based on Lemma 3.4, for any fixed $r > 0$, the real parts of any sequence of characteristic roots with unbounded modulus must tend to $-\infty$. Therefore, regardless of whether there exist complex roots with positive real part, complex roots with negative real part are always present. Thus ϕ does not converge to u_- .

In Lemma 3.1, there are three further cases, not yet mentioned, in which equation (2.6) admits a positive real root:

$$\begin{cases} s = f'(u_-) < 0, & r > \hat{r}_1, \\ s < f'(u_-) < 0, \\ s < 0 < f'(u_-) < -s. \end{cases}$$

All of the above situations yield $|k_2^{(2)}| \leq |k_1^{(2)}|$, then the real parts of all complex roots of equation (2.6) are negative, which implies that ϕ fails to converge to u_- . Thus, the proof of Proposition 3.1 is complete. $\square$

Proof of Proposition 3.2. For $m = 0$, we formally have

$$\lim_{\xi \rightarrow \pm\infty} \frac{\phi_\xi}{\phi} = 0,$$

integrating the first equations of (2.2) from $-\infty$ to $+\infty$, then we obtain

$$-s\phi \Big|_{-\infty}^{+\infty} - D \frac{\phi_\xi}{\phi} \Big|_{-\infty}^{+\infty} + f(\phi(\xi + sr)) \Big|_{-\infty}^{+\infty} = 0,$$

thus s satisfies the Rankine-Hugoniot condition (2.4). Integrating (2.2) from $-\infty$ to ξ and ξ to $+\infty$, we have

$$-s(\phi - u_{\pm}) - D \frac{\phi_{\xi}}{\phi} + f(\phi(\xi + sr)) - f(u_{\pm}) = 0. \quad (3.24)$$

Rewrite the equality we will see

$$\phi_{\xi} = \frac{1}{D} \phi [f(\phi(\xi + sr)) - f(u_{\pm}) - s(\phi - u_{\pm})]. \quad (3.25)$$

If $f'(0) \neq s$,

$$\begin{aligned} \phi_{\xi} &= \frac{1}{D} \phi [f(\phi(\xi + sr)) - f(0) - s(\phi - 0)] \\ &= \frac{1}{D} \phi^2 \left[\frac{f(\phi(\xi + sr)) - f(0)}{\phi - 0} - s \right], \end{aligned} \quad (3.26)$$

thus we can derive that

$$\phi = O(1) \frac{1}{|\xi|}, \quad \text{as } \xi \rightarrow +\infty,$$

consistent with the preceding analysis when $0 < m < 1$.

If $f'(0) = s$, by Taylor's expansion, we deduce that

$$\begin{aligned} \phi_{\xi} &= \frac{1}{D} \phi [f(\phi(\xi + sr)) - f(0) - s(\phi - 0)] \\ &= \frac{1}{D} \phi \left[f'(0)(\phi(\xi + sr) - \phi(\xi)) + \frac{f^{(k+1)}(0)}{(k+1)!} \phi^{k+1}(\xi) \frac{\phi^{k+1}(\xi + sr)}{\phi^{k+1}(\xi)} + o(\phi^{k+2}(\xi + sr)) \right]. \end{aligned}$$

Similarly, we can obtain the asymptotic behavior around $u = 0$,

$$\phi = O(1) \frac{1}{|\xi|^{\frac{1}{k+1}}}, \quad \text{as } \xi \rightarrow +\infty, \quad f'(0) = s.$$

Next we linearize equation (3.26) around u_{-} ,

$$\phi_{\xi} + \frac{su_{-}}{D}(\phi - u_{-}) = \frac{f'(u_{-})u_{-}^{1-m}}{D}(\phi(\xi + sr) - u_{-}). \quad (3.27)$$

Since (3.27) is replaced $1 - m$ with 1 in (3.21), thus we omit the details about the derivation of oscillatory. $\square$

Proof of Proposition 3.3. For the case where $m < 0$, formally we have

$$\lim_{\xi \rightarrow \pm\infty} \frac{\phi_{\xi}}{\phi^{1+|m|}} = 0,$$

integrating the first equation of (2.3) from $-\infty$ to $+\infty$, then we obtain

$$-s\phi \Big|_{-\infty}^{+\infty} - D \frac{\phi_{\xi}}{\phi^{1+|m|}} \Big|_{-\infty}^{+\infty} + f(\phi(\xi + sr)) \Big|_{-\infty}^{+\infty} = 0,$$

thus $s = \frac{f(u_{-}) - f(0)}{u_{-} - 0}$. Integrating (2.3) from $-\infty$ to ξ and from ξ to $+\infty$, then we obtain

$$-s(\phi - u_{\pm}) - D \frac{\phi_{\xi}}{\phi^{1+|m|}} + f(\phi(\xi + sr)) - f(u_{\pm}) = 0, \quad (3.28)$$

which implies that

$$\phi_{\xi} = \frac{1}{D} \phi^{1+|m|} [f(\phi(\xi + sr)) - f(u_{\pm}) - s(\phi - u_{\pm})]. \quad (3.29)$$

The asymptotic behavior of $m < 0$ as $\xi \rightarrow \pm\infty$ remain identical to that of $0 < m < 1$, here we omit the details. $\square$

4 Existence and asymptotic behavior of the travelling waves

In this section, we establish the existence of travelling wave solutions to equations (1.16)-(1.18). Motivated by the asymptotic analysis in Section 3, which provides expected decay rates as $\xi \rightarrow \pm\infty$, we introduce suitably weighted function spaces in which the Leray-Schauder fixed-point theorem delivers a solution.

We then turn to the asymptotic behavior and oscillatory of travelling waves. By applying the asymptotic analysis developed in Section 3 to the wave profiles, we rigorously verify the predicted exponential rates, including the critical parameter boundary between monotonic and oscillatory convergence.

First of all, let us recall three well-known results as follows.

Lemma 4.1 Suppose $a, b \in \mathbb{R}$ and $p > 1$, it holds that

$$|a - b|^p \leq 2^{p-1}(|a|^p + |b|^p). \quad (4.1)$$

Lemma 4.2 Suppose $a \in \mathbb{R}, b > 0$ and $0 < p < 1$, if $a + b > 0$, it holds that

$$(a + b)^p \leq |a|^p + b^p. \quad (4.2)$$

Lemma 4.3 (Leray-Schauder Theorem) Suppose $T : E \rightarrow E$ is completely continuous. If $\{x \mid x \in E, x = \theta Tx, 0 < \theta < 1\}$ is bounded, then T must have a fixed point in the closed sphere $B(0, R)$ of E , where

$$R = \sup \{\|x\| \mid x = \theta Tx, 0 < \theta < 1\}.$$

Proof of Theorem 2.1. For $0 < m < 1$, we define a smooth function

$$\ell_1(\xi) = u_- - u_- \frac{1}{1 + e^{-\Lambda_1 \xi}},$$

which satisfies $\ell_1(\pm\infty) = u_{\pm}$ with $u_- > u_+ = 0$, where $\Lambda_1 = \min\{\lambda_1, \nu_1\} > 0$. Define $\psi = \phi - \ell_1$, then we have

$$\begin{cases} \psi_{\xi} = \frac{1}{D}(\psi + \ell_1)^{1-m} [f((\psi + \ell_1)(\xi + sr)) - f(u_{\pm}) - s(\psi + \ell_1 - u_{\pm})] - \ell_{1\xi}, \\ \psi(\pm\infty) = 0, \end{cases} \quad (4.3)$$

the equation of ψ can be deduced by equation (3.18) and $\phi(\pm\infty) = u_{\pm}$. The asymptotic behavior of $\ell_1(\xi)$ is easy to obtained by the definition of $\ell_1(\xi)$,

$$|\ell_1(\xi)| = u_- \left(1 - \frac{1}{1 + e^{-\Lambda_1 \xi}}\right) = u_- \frac{1}{1 + e^{\Lambda_1 \xi}} \leq C e^{-\Lambda_1 |\xi|}, \quad \text{when } \xi \geq 0, \quad (4.4)$$

and

$$|\ell_1(\xi) - u_-| = u_- \frac{1}{1 + e^{-\Lambda_1 \xi}} \leq C e^{-\Lambda_1 |\xi|}, \quad \text{when } \xi \leq 0, \quad (4.5)$$

and

$$|\ell_{1\xi}| = \left| -u_- \Lambda_1 \frac{e^{\Lambda_1 \xi}}{(1 + e^{\Lambda_1 \xi})^2} \right| \leq C e^{-\Lambda_1 |\xi|}, \quad \xi \in (-\infty, +\infty). \quad (4.6)$$

Recalling the analysis presented in Proposition 3.1, we can deduce that if $f'(0) < s \leq f'(u_-)$,

$$|\psi| \leq |\phi| + |\ell_1| \leq C\xi^{-\frac{1}{1-m}} + Ce^{-\Lambda_1|\xi|} \leq C\xi^{-\frac{1}{1-m}}, \quad \text{as } \xi \rightarrow +\infty, \quad (4.7)$$

and if $f'(0) = s < f'(u_-)$,

$$|\psi| \leq |\phi| + |\ell_1| \leq C\xi^{-\frac{1}{k+1-m}} + Ce^{-\Lambda_1|\xi|} \leq C\xi^{-\frac{1}{k+1-m}}, \quad \text{as } \xi \rightarrow +\infty, \quad (4.8)$$

and

$$|\psi| = |\phi - u_- - (\ell_1 - u_-)| \leq |\phi - u_-| + |\ell_1 - u_-| \leq Ce^{-\Lambda_1|\xi|}, \quad \text{as } \xi \rightarrow -\infty, \quad (4.9)$$

thus ψ converges algebraically to 0 at around $+\infty$ and converges exponentially to 0 at around $-\infty$. Define a weighted norm space

$$X = \left\{ \psi(\xi) \in C(\mathbb{R}) \mid \|\psi\|_X < \infty, \psi \sim C_+ \xi^{-\frac{1}{k+1-m}} (\xi \rightarrow +\infty), \psi \sim C_- e^{-\Lambda_1|\xi|} (\xi \rightarrow -\infty) \right\},$$

and the weighted norm

$$\|\psi\|_X = \max \left\{ \sup_{\xi \geq 0} (1 + \xi)^{\alpha_1} |\psi(\xi)|, \sup_{\xi \leq 0} e^{\beta_1|\xi|} |\psi(\xi)| \right\},$$

where $0 < \alpha_1 < \frac{1}{k+1-m}$ and $0 < \beta_1 < \Lambda_1$.

Denote $\psi_\xi \triangleq h_1(\xi, \psi(\xi)) - \ell_{1\xi}$, then from (4.3) it follows that

$$\begin{cases} \psi_\xi + \mu\psi(\xi) = h_1(\xi, \psi(\xi)) - \ell_{1\xi} + \mu\psi(\xi), & \mu > 0, \\ \psi(\pm\infty) = 0. \end{cases} \quad (4.10)$$

Taking a kernel function

$$G(\xi, \eta) = \begin{cases} e^{-\mu(\xi-\eta)}, & \xi \geq \eta, \\ 0, & \xi < \eta, \end{cases} \quad (4.11)$$

and considering the following integral equation

$$\begin{aligned} \psi(\xi) &= \int_{-\infty}^{\infty} G(\xi, \eta) [h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)] d\eta \\ &= \int_{-\infty}^{\xi} e^{-\mu(\xi-\eta)} [h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)] d\eta. \end{aligned} \quad (4.12)$$

Since ψ converges algebraically to 0 as $\xi \rightarrow +\infty$ and converges exponentially to 0 as $\xi \rightarrow -\infty$, we can deduce that

$$|\psi(\xi)| \leq Ce^{-\mu\xi} + C\xi(e^{-\frac{\mu}{2}\xi} + e^{-\frac{\Lambda_1}{2}\xi}) + C(1 + \xi)^{-\frac{1}{1-m}}, \quad f'(0) < s \leq f'(u_-), \quad \xi \geq 0,$$

$$|\psi(\xi)| \leq Ce^{-\mu\xi} + C\xi(e^{-\frac{\mu}{2}\xi} + e^{-\frac{\Lambda_1}{2}\xi}) + C(1 + \xi)^{-\frac{1}{k+1-m}}, \quad f'(0) = s < f'(u_-), \quad \xi \geq 0,$$

and

$$|\psi(\xi)| \leq Ce^{-\Lambda_1|\xi|}, \quad \xi \leq 0,$$

from the subsequent estimation of $h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)$. In either case the estimates imply $\psi(\xi) \rightarrow 0$ as $\xi \rightarrow \pm\infty$, which is consistent with the prescribed boundary conditions in (4.10). Hence, (4.12) is equivalent to (4.10).

Basing on (4.12), we define

$$T_1\psi(\xi) = \int_{-\infty}^{\xi} e^{-\mu(\xi-\eta)} \left[h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta) \right] d\eta,$$

we are going to verify that T_1 satisfies the condition of Leray-Schauder theorem.

Firstly we need to make sure that $T_1\psi \in X$. It follows easily from inequalities (4.7)–(4.9) that for all $\psi \in X$, when $\xi \leq 0$,

$$|\psi| \leq Ce^{-\Lambda_1|\xi|}, \quad (4.13)$$

while for $\xi \geq 0$,

$$|\psi| \leq C(1 + \xi)^{-\frac{1}{1-m}}, \quad \text{if } f'(0) < s \leq f'(u_-), \quad (4.14)$$

or

$$|\psi| \leq C(1 + \xi)^{-\frac{1}{k+1-m}}, \quad \text{if } f'(0) = s < f'(u_-). \quad (4.15)$$

We are going to estimate h_1 , when $\eta \leq 0$,

$$\begin{aligned} |h_1(\eta, \psi(\eta))| &= \frac{1}{D} \left| (\psi + \ell_1)^{1-m} [f(\psi + \ell_1)(\eta + sr) - f(u_-) - s(\psi + \ell_1 - u_-)] \right| \\ &\leq \frac{1}{D} \left| \left(|\psi|^{1-m} + \ell_1^{1-m} \right) (\psi + \ell_1 - u_-) \left[\frac{f(\psi + \ell_1)(\eta + sr) - f(u_-)}{\psi + \ell_1 - u_-} - s \right] \right| \\ &\leq C \left(|\psi|^{2-m} + |\psi|^{1-m} |\ell_1 - u_-| + \ell_1^{1-m} |\psi| + \ell_1^{1-m} |\ell_1 - u_-| \right) \\ &\leq C \left(e^{-(2-m)\Lambda_1|\eta|} + e^{-(1-m)\Lambda_1|\eta|} \cdot \frac{1}{1 + e^{\Lambda_1\eta}} + u_-^{1-m} e^{-\Lambda_1|\eta|} + \frac{u_-^{1-m}}{1 + e^{\Lambda_1\eta}} \right) \\ &\leq C \left(e^{-(2-m)\Lambda_1|\eta|} + e^{-\Lambda_1|\eta|} \right) \\ &\leq Ce^{-\Lambda_1|\eta|}, \end{aligned} \quad (4.16)$$

where we have utilized Lemma 4.2. Applying Lemma 4.1, when $\eta \geq 0$, if $f'(0) < s \leq f'(u_-)$, $k = 0$,

$$\begin{aligned} |h_1(\eta, \psi(\eta))| &= \frac{1}{D} \left| (\psi + \ell_1)^{1-m} [f(\psi + \ell_1)(\eta + sr) - f(0) - s(\psi + \ell_1)] \right| \\ &= \frac{1}{D} \left| (\psi + \ell_1)^{2-m} \left[\frac{f(\psi + \ell_1)(\eta + sr) - f(0)}{\psi + \ell_1 - 0} - s \right] \right| \\ &\leq C |\psi + \ell_1|^{2-m} \\ &\leq C 2^{1-m} (|\psi|^{2-m} + |\ell_1|^{2-m}) \\ &\leq C \left[(1 + \eta)^{-1-\frac{1}{1-m}} + u_-^{2-m} \left(\frac{1}{1 + e^{\Lambda_1|\eta|}} \right)^{2-m} \right] \\ &\leq C \left[(1 + \eta)^{-1-\frac{1}{1-m}} + e^{-(2-m)\Lambda_1|\eta|} \right] \\ &\leq C(1 + \eta)^{-1-\frac{1}{1-m}} + e^{-\Lambda_1|\eta|}, \end{aligned} \quad (4.17)$$

or if $f'(0) = s < f'(u_-)$, $k \geq 1$,

$$|h_1(\eta, \psi(\eta))| = \frac{1}{D} \left| (\psi + \ell_1)^{1-m} \left[f'(0)(\psi + \ell_1)(\eta + sr) - s(\psi + \ell_1)(\eta) \right] \right|$$

$$\begin{aligned}
& \left| \frac{f^{(k+1)}(0)}{(k+1)!} (\psi + \ell_1)^{k+1} (\eta + sr) + o\left((\psi + \ell_1)^{k+2} (\eta + sr)\right) \right| \\
& \leq C |\psi + \ell_1|^{1-m} \left(|\psi + \ell_1| + |\psi + \ell_1|^{k+1} \right) \\
& \leq C |\psi + \ell_1|^{2-m} + C |\psi + \ell_1|^{k+2-m} \\
& \leq C \left(|\psi|^{2-m} + |\psi|^{k+2-m} + |\ell_1|^{2-m} + |\ell_1|^{k+2-m} \right) \\
& \leq C \left[(1 + \eta)^{-\frac{2-m}{k+1-m}} + (1 + \eta)^{-1-\frac{1}{k+1-m}} + e^{-(2-m)\Lambda_1|\eta|} + e^{-(k+2-m)\Lambda_1|\eta|} \right] \\
& \leq C(1 + \eta)^{-\frac{2-m}{k+1-m}} + C(1 + \eta)^{-1-\frac{1}{k+1-m}} + e^{-\Lambda_1|\eta|}. \tag{4.18}
\end{aligned}$$

When $\xi \leq 0$, according to (4.16), (4.6) and (4.13), we derive that

$$\begin{aligned}
|T_1\psi(\xi)| &= \left| \int_{-\infty}^{\xi} e^{-\mu(\xi-\eta)} \left[h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta) \right] d\eta \right| \\
&\leq C e^{-\mu\xi} \int_{-\infty}^{\xi} e^{-\mu|\eta|} e^{-\Lambda_1|\eta|} d\eta \\
&\leq C e^{-\mu\xi} e^{-(\mu+\Lambda_1)|\xi|} \\
&\leq C e^{-\Lambda_1|\xi|},
\end{aligned}$$

and

$$e^{\beta_1|\xi|} |T_1\psi(\xi)| \leq C e^{-(\Lambda_1-\beta_1)|\xi|}. \tag{4.19}$$

When $\xi \geq 0$, we divide $(-\infty, \xi)$ into $((-\infty, 0))$, $(0, \frac{\xi}{2})$ and $(\frac{\xi}{2}, \xi)$ and employing the estimations (4.16)- (4.18) to estimate the integral. When $\eta \leq 0$,

$$\begin{aligned}
& \left| \int_{-\infty}^0 e^{-\mu(\xi-\eta)} \left[h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta) \right] d\eta \right| \\
& \leq C e^{-\mu\xi} \int_{-\infty}^0 e^{-(\mu+\Lambda_1)|\eta|} d\eta \\
& \leq C e^{-\mu\xi}.
\end{aligned}$$

If $f'(0) < s \leq f'(u_-)$, when $0 < \eta < \frac{\xi}{2}$, since $\mu > 0$, we have $e^{-\mu(\xi-\eta)} < e^{-\frac{\mu}{2}\xi}$, then by (4.16), (4.6) and (4.14),

$$\begin{aligned}
& \left| \int_0^{\frac{\xi}{2}} e^{-\mu(\xi-\eta)} \left[h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta) \right] d\eta \right| \\
& \leq C e^{-\frac{\mu}{2}\xi} \int_0^{\frac{\xi}{2}} \left[(1 + \eta)^{-1-\frac{1}{1-m}} + (1 + \eta)^{-\frac{1}{1-m}} + e^{-\Lambda_1|\eta|} \right] d\eta \\
& \leq C e^{-\frac{\mu}{2}\xi} \int_0^{\frac{\xi}{2}} 3 d\eta \\
& \leq C \xi e^{-\frac{\mu}{2}\xi},
\end{aligned}$$

when $\frac{\xi}{2} < \eta < \xi$,

$$(1 + \eta)^{-1-\frac{1}{1-m}} < (1 + \eta)^{-\frac{1}{1-m}} < \left(1 + \frac{\xi}{2}\right)^{-\frac{1}{1-m}} < 2^{\frac{1}{1-m}} (1 + \xi)^{-\frac{1}{1-m}},$$

thus we obtain that

$$\begin{aligned}
& \left| \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} [h(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)] d\eta \right| \\
& \leq C \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} \left[(1+\eta)^{-1-\frac{1}{1-m}} + (1+\eta)^{-\frac{1}{1-m}} + e^{-\Lambda_1|\eta|} \right] d\eta \\
& \leq C \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} \left[(1+\eta)^{-\frac{1}{1-m}} + e^{-\Lambda_1|\eta|} \right] d\eta \\
& \leq C(1+\xi)^{-\frac{1}{1-m}} \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} d\eta + C \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} e^{-\Lambda_1|\eta|} d\eta \\
& \leq \frac{C}{\mu} (1+\xi)^{-\frac{1}{1-m}} \left(1 - e^{-\mu\frac{\xi}{2}} \right) + C \int_{\frac{\xi}{2}}^{\xi} e^{-\frac{\Lambda_1}{2}\xi} d\eta \\
& \leq C(1+\xi)^{-\frac{1}{1-m}} + C\xi e^{-\frac{\Lambda_1}{2}\xi}.
\end{aligned}$$

Combine the above three integrals,

$$\begin{aligned}
|T_1\psi(\xi)| &= \left| \int_{-\infty}^{\xi} e^{-\mu(\xi-\eta)} [h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)] d\eta \right| \\
&\leq C e^{-\mu\xi} + C\xi(e^{-\frac{\mu}{2}\xi} + e^{-\frac{\Lambda_1}{2}\xi}) + C(1+\xi)^{-\frac{1}{1-m}}, \quad \xi \geq 0,
\end{aligned}$$

hence we have

$$\begin{aligned}
& (1+\xi)^{\alpha_1} |T_1\psi(\xi)| \\
& \leq C(1+\xi)^{\alpha_1} e^{-\mu\xi} + C\xi(1+\xi)^{\alpha_1} (e^{-\frac{\mu}{2}\xi} + e^{-\frac{\Lambda_1}{2}\xi}) + C(1+\xi)^{-(\frac{1}{1-m}-\alpha_1)} \\
& \leq C(1+\xi)^{-(\frac{1}{1-m}-\alpha_1)}, \quad \xi \geq 0.
\end{aligned} \tag{4.20}$$

If $f'(0) = s < f'(u_-)$, when $0 < \eta < \frac{\xi}{2}$,

$$\begin{aligned}
& \left| \int_0^{\frac{\xi}{2}} e^{-\mu(\xi-\eta)} [h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)] d\eta \right| \\
& \leq C e^{-\frac{\mu}{2}\xi} \int_0^{\frac{\xi}{2}} \left[(1+\eta)^{-\frac{2-m}{k+1-m}} + (1+\eta)^{-1-\frac{1}{k+1-m}} + (1+\eta)^{-\frac{1}{k+1-m}} + e^{-\Lambda_1|\eta|} \right] d\eta \\
& \leq C e^{-\frac{\mu}{2}\xi} \int_0^{\frac{\xi}{2}} 4 d\eta \\
& \leq C\xi e^{-\frac{\mu}{2}\xi},
\end{aligned}$$

when $\frac{\xi}{2} < \eta < \xi$,

$$(1+\eta)^{-1-\frac{1}{k+1-m}} < (1+\eta)^{-\frac{2-m}{k+1-m}} < (1+\eta)^{-\frac{1}{k+1-m}} < \left(1+\frac{\xi}{2}\right)^{-\frac{1}{k+1-m}} < 2^{\frac{1}{k+1-m}} (1+\xi)^{-\frac{1}{k+1-m}},$$

since $\eta \in (\frac{\xi}{2}, \xi)$, the above inequality yields that

$$\left| \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} [h(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)] d\eta \right|$$

$$\begin{aligned}
&\leq C \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} \left[(1+\eta)^{-\frac{2-m}{k+1-m}} + (1+\eta)^{-1-\frac{1}{k+1-m}} + (1+\eta)^{-\frac{1}{k+1-m}} + e^{-\Lambda_1|\eta|} \right] d\eta \\
&\leq C \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} \left[(1+\eta)^{-\frac{1}{k+1-m}} + e^{-\Lambda_1|\eta|} \right] d\eta \\
&\leq C(1+\xi)^{-\frac{1}{k+1-m}} \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} d\eta + C\xi e^{-\frac{\Lambda_1}{2}\xi} \\
&= \frac{C}{\mu} (1+\xi)^{-\frac{1}{k+1-m}} \left(1 - e^{-\mu\frac{\xi}{2}} \right) + C\xi e^{-\frac{\Lambda_1}{2}\xi} \\
&\leq C(1+\xi)^{-\frac{1}{k+1-m}} + C\xi e^{-\frac{\Lambda_1}{2}\xi}.
\end{aligned}$$

Then we deduce that

$$\begin{aligned}
|T_1\psi(\xi)| &= \left| \int_{-\infty}^{\xi} e^{-\mu(\xi-\eta)} \left[h_1(\eta, \psi(\eta)) - \ell_1\eta + \mu\psi(\eta) \right] d\eta \right| \\
&\leq C e^{-\mu\xi} + C\xi (e^{-\frac{\mu}{2}\xi} + e^{-\frac{\Lambda_1}{2}\xi}) + C(1+\xi)^{-\frac{1}{k+1-m}}, \quad \xi \geq 0,
\end{aligned}$$

and

$$\begin{aligned}
&(1+\xi)^{\alpha_1} |T_1\psi(\xi)| \\
&\leq C(1+\xi)^{\alpha_1} e^{-\mu\xi} + C\xi(1+\xi)^{\alpha_1} (e^{-\frac{\mu}{2}\xi} + e^{-\frac{\Lambda_1}{2}\xi}) + C(1+\xi)^{-(\frac{1}{k+1-m}-\alpha_1)} \\
&\leq C(1+\xi)^{-(\frac{1}{k+1-m}-\alpha_1)}, \quad \xi \geq 0.
\end{aligned} \tag{4.21}$$

By the virtue of (4.19)-(4.21), we obtained that $\|T_1\psi\| \leq C$, $T_1\psi$ converges exponentially to 0 at around $-\infty$ and converges algebraically to 0 at around $+\infty$, thus $T_1\psi \in X$.

Next we are going to verify that T_1 is completely continuous. Letting $\psi_n, \psi \in X$, $\|\psi_n - \psi\|_X \rightarrow 0$ ($n \rightarrow \infty$), we have

$$|T_1\psi_n - T_1\psi| = \left| \int_{-\infty}^{\xi} e^{-\mu(\xi-\eta)} \left[h_1(\eta, \psi_n(\eta)) - h_1(\eta, \psi(\eta)) + \mu(\psi_n - \psi) \right] d\eta \right|.$$

When $\eta \leq 0$, since ψ satisfies (4.13), we calculate that

$$\begin{aligned}
&|h_1(\eta, \psi_n(\eta)) - h_1(\eta, \psi(\eta))| \\
&= \frac{1}{D} \left| (\psi_n + \ell_1)^{1-m} [f(\psi_n + \ell_1)(\eta + sr) - f(u_-) - s(\psi_n + \ell_1 - u_-)] \right. \\
&\quad \left. - (\psi + \ell_1)^{1-m} [f(\psi + \ell_1)(\eta + sr) - f(u_-) - s(\psi + \ell_1 - u_-)] \right| \\
&= \frac{1}{D} \left| [(\psi_n + \ell_1)^{1-m} - (\psi + \ell_1)^{1-m}] (\psi_n + \ell_1 - u_-) \left[\frac{f(\psi_n + \ell_1)(\eta + sr) - f(u_-)}{\psi_n + \ell_1 - u_-} - s \right] \right. \\
&\quad \left. + (\psi + \ell_1)^{1-m} [(f(\psi_n + \ell_1) - f(\psi + \ell_1))(\eta + sr) - s(\psi_n - \psi)] \right| \\
&\leq C \frac{1}{D} |(\psi_n - \psi) \min \{(\psi + \ell_1)^{-m}, (\psi_n + \ell_1)^{-m}\} (\psi_n + \ell_1 - u_-)| \\
&\quad + \frac{1}{D} \left| (\psi^{1-m} + \ell_1^{1-m})(\psi_n - \psi) \left[\frac{(f(\psi_n + \ell_1) - f(\psi + \ell_1))(\eta + sr)}{(\psi_n + \ell_1) - (\psi + \ell_1)} - s \right] \right| \\
&\leq C|\psi_n - \psi| (|\psi_n| + |\ell_1 - u_-| + |\psi|^{1-m} + |\ell_1|^{1-m}) \\
&\leq C|\psi_n - \psi| \left(e^{-\Lambda_1|\eta|} + \frac{1}{1 + e^{-\Lambda_1\eta}} + e^{-(1-m)\Lambda_1|\eta|} + u_-^{1-m} \right)
\end{aligned}$$

$$\leq C|\psi_n - \psi| \left(e^{-(1-m)\Lambda_1|\eta|} + e^{-\Lambda_1|\eta|} + 1 \right), \quad (4.22)$$

where the first inequality is obtained by applying the Mean Value theorem. When $\eta \geq 0$, if $f'(0) < s \leq f'(u_-)$, $k = 0$, then

$$\begin{aligned} & |h_1(\eta, \psi_n(\eta)) - h_1(\eta, \psi(\eta))| \\ &= \frac{1}{D} \left| (\psi_n + \ell_1)^{1-m} [f(\psi_n + \ell_1)(\eta + sr) - f(0) - s(\psi_n + \ell_1 - 0)] \right. \\ &\quad \left. - (\psi + \ell_1)^{1-m} [f(\psi + \ell_1)(\eta + sr) - f(0) - s(\psi + \ell_1 - 0)] \right| \\ &= \frac{1}{D} \left| \left[(\psi_n + \ell_1)^{1-m} - (\psi + \ell_1)^{1-m} \right] (\psi_n + \ell_1) \left[\frac{f(\psi_n + \ell_1)(\eta + sr) - f(0)}{\psi_n + \ell_1 - 0} - s \right] \right. \\ &\quad \left. + (\psi + \ell_1)^{1-m} [(f(\psi_n + \ell_1) - f(\psi + \ell_1))(\eta + sr) - s(\psi_n - \psi)] \right| \\ &\leq C \frac{1}{D} \left| (\psi_n - \psi) \min \{ (\psi + \ell_1)^{-m}, (\psi_n + \ell_1)^{-m} \} (\psi_n + \ell_1) \right| \\ &\quad + \frac{1}{D} \left| (\psi^{1-m} + \ell_1^{1-m})(\psi_n - \psi) \left[\frac{(f(\psi_n + \ell_1) - f(\psi + \ell_1))(\eta + sr)}{(\psi_n + \ell_1) - (\psi + \ell_1)} - s \right] \right| \\ &\leq C|\psi_n - \psi| \left(|\psi_n + \ell_1|^{1-m} + |\psi|^{1-m} + |\ell_1|^{1-m} \right) \\ &\leq C|\psi_n - \psi| \left(|\psi_n|^{1-m} + |\psi|^{1-m} + |\ell_1|^{1-m} \right) \\ &\leq C|\psi_n - \psi| \left[(1 + \eta)^{-1} + e^{-(1-m)\Lambda_1|\eta|} \right], \end{aligned} \quad (4.23)$$

where we have employed (4.14). If $f'(0) = s < f'(u_-)$, $k \geq 1$, utilizing the Taylor expansion and (4.15), we deduce that

$$\begin{aligned} & |h_1(\eta, \psi_n(\eta)) - h_1(\eta, \psi(\eta))| \\ &\leq \frac{1-m}{D} \left| (\psi_n - \psi) \min \{ (\psi + \ell_1)^{-m}, (\psi_n + \ell_1)^{-m} \} \left[f'(0)(\psi_n + \ell_1)(\eta + sr) - s(\psi_n + \ell_1) \right. \right. \\ &\quad \left. \left. + \frac{f^{(k+1)}(0)}{(k+1)!} (\psi_n + \ell_1)^{k+1} (\eta + sr) + o((\psi_n + \ell_1)^{k+1} (\eta + sr)) \right] \right| \\ &\quad + \frac{1}{D} \left| (\psi^{1-m} + \ell_1^{1-m}) \left[f'(0)(\psi_n - \psi)(\eta + sr) - s(\psi_n - \psi) \right. \right. \\ &\quad \left. \left. + \frac{f^{(k+1)}(0)}{(k+1)!} (\psi_n^{k+1} (\eta + sr) - \psi^{k+1} (\eta + sr)) + o(\psi_n^{k+2} (\eta + sr)) + o(\psi^{k+2} (\eta + sr)) \right] \right| \\ &\leq C|\psi_n - \psi| \left(|\psi_n + \ell_1|^{1-m} + |\psi_n + \ell_1|^{k+1-m} (\eta + sr) \right) \\ &\quad + C(|\psi|^{1-m} + |\ell_1|^{1-m}) \left(|\psi_n - \psi| + |\psi_n^{k+1} - \psi^{k+1}| (\eta + sr) \right) \\ &\leq C|\psi_n - \psi| \left(|\psi_n|^{1-m} + |\psi_n|^{k+1-m} + |\ell_1|^{1-m} + |\ell_1|^{k+1-m} \right) \\ &\quad + C(|\psi|^{1-m} + |\ell_1|^{1-m}) \left(|\psi_n - \psi| + |\psi_n - \psi|^{k+1} \right) \\ &\leq C|\psi_n - \psi| \left(|\psi_n|^{1-m} + |\psi_n|^{k+1-m} + |\psi|^{1-m} + |\psi_n - \psi|^k + |\ell_1|^{1-m} + |\ell_1|^{k+1-m} \right) \\ &\leq C|\psi_n - \psi| \left[(1 + \eta)^{-1} + (1 + \eta)^{-\frac{1-m}{k+1-m}} + (1 + \eta)^{-\frac{k}{k+1-m}} + e^{-(1-m)\Lambda_1|\eta|} + e^{-(k+1-m)\Lambda_1|\eta|} \right]. \end{aligned} \quad (4.24)$$

When $\xi \leq 0$, recalling (4.22),

$$e^{\beta_1|\xi|} |T_1\psi_n - T_1\psi|$$

$$\begin{aligned}
&= e^{\beta_1|\xi|} \left| \int_{-\infty}^{\xi} e^{-\mu(\xi-\eta)} [h_1(\eta, \psi_n(\eta)) - h_1(\eta, \psi(\eta)) + \mu(\psi_n - \psi)] d\eta \right| \\
&\leq C e^{\beta_1|\xi|} e^{-\mu\xi} \int_{-\infty}^{\xi} e^{-\mu|\eta|} |\psi_n - \psi| \left(e^{-(1-m)\Lambda_1|\eta|} + e^{-\Lambda_1|\eta|} + 1 \right) d\eta \\
&\leq C e^{\beta_1|\xi|} e^{-\mu\xi} \int_{-\infty}^{\xi} e^{-\mu|\eta|} e^{\Lambda_1|\eta|} |\psi_n - \psi| \left(e^{-(2-m)\Lambda_1|\eta|} + e^{-2\Lambda_1|\eta|} + e^{-\Lambda_1|\eta|} \right) d\eta \\
&\leq C e^{\beta_1|\xi|} e^{-\mu\xi} \|\psi_n - \psi\|_X \int_{-\infty}^{\xi} \left(e^{-((2-m)\Lambda_1+\mu)|\eta|} + e^{-(2\Lambda_1+\mu)|\eta|} + e^{-(\Lambda_1+\mu)|\eta|} \right) d\eta \\
&\leq C e^{\beta_1|\xi|} \left(e^{-(2-m)\Lambda_1|\xi|} + e^{-2\Lambda_1|\xi|} + e^{-\Lambda_1|\xi|} \right) \|\psi_n - \psi\|_X \\
&\leq C e^{-(\Lambda_1-\beta_1)|\xi|} \|\psi_n - \psi\|_X.
\end{aligned}$$

When $\xi \geq 0$, we estimate the integral by utilizing (4.22)-(4.24). Since $\eta \in (-\infty, 0)$,

$$\begin{aligned}
&\left| \int_{-\infty}^0 e^{-\mu(\xi-\eta)} [h_1(\eta, \psi_n(\eta)) - h_1(\eta, \psi(\eta)) + \mu(\psi_n - \psi)] d\eta \right| \\
&\leq C e^{-\mu\xi} \int_{-\infty}^0 e^{-\mu|\eta|} |\psi_n - \psi| \left(e^{-(1-m)\Lambda_1|\eta|} + e^{-\Lambda_1|\eta|} + 1 \right) d\eta \\
&\leq C e^{-\mu\xi} \int_{-\infty}^0 e^{-\mu|\eta|} e^{\Lambda_1|\eta|} |\psi_n - \psi| e^{-\Lambda_1|\eta|} d\eta \\
&\leq C e^{-\mu\xi} \|\psi_n - \psi\|_X \int_{-\infty}^0 e^{-(\mu+\Lambda_1)|\eta|} d\eta \\
&\leq C e^{-\mu\xi} \|\psi_n - \psi\|_X.
\end{aligned}$$

If $f'(0) < s \leq f'(u_-)$, $k = 0$, when $0 < \eta < \frac{\xi}{2}$, $(1 + \eta)^{-1} \leq 1$ and $e^{-(1-m)\Lambda_1|\eta|} \leq 1$, then

$$\begin{aligned}
&\left| \int_0^{\frac{\xi}{2}} e^{-\mu(\xi-\eta)} [h_1(\eta, \psi_n(\eta)) - h_1(\eta, \psi(\eta)) + \mu(\psi_n - \psi)] d\eta \right| \\
&\leq C e^{-\frac{\mu}{2}\xi} \int_0^{\frac{\xi}{2}} |\psi_n - \psi| \left[(1 + \eta)^{-1} + e^{-(1-m)\Lambda_1|\eta|} + 1 \right] d\eta \\
&\leq C e^{-\frac{\mu}{2}\xi} \int_0^{\frac{\xi}{2}} (1 + \eta)^{\frac{1}{1-m}} |\psi_n - \psi| (1 + \eta)^{-\frac{1}{1-m}} d\eta \\
&\leq C e^{-\frac{\mu}{2}\xi} \|\psi_n - \psi\|_X \int_0^{\frac{\xi}{2}} (1 + \eta)^{-\frac{1}{1-m}} d\eta \\
&\leq C \xi e^{-\frac{\mu}{2}\xi} \|\psi_n - \psi\|_X,
\end{aligned}$$

when $\frac{\xi}{2} < \eta < \xi$, $(1 + \eta)^{-1} \leq 1$ and $e^{-(1-m)\Lambda_1|\eta|} \leq 1$,

$$\begin{aligned}
&\left| \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} [h_1(\eta, \psi_n(\eta)) - h_1(\eta, \psi(\eta)) + \mu(\psi_n - \psi)] d\eta \right| \\
&\leq \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} |\psi_n - \psi| \left[(1 + \eta)^{-1} + e^{-(1-m)\Lambda_1|\eta|} + 1 \right] d\eta \\
&\leq \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} (1 + \eta)^{\frac{1}{1-m}} |\psi_n - \psi| (1 + \eta)^{-\frac{1}{1-m}} d\eta
\end{aligned}$$

$$\begin{aligned}
&\leq C(1+\xi)^{-\frac{1}{1-m}} \|\psi_n - \psi\|_X \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} d\eta \\
&\leq C(1+\xi)^{-\frac{1}{1-m}} \|\psi_n - \psi\|_X,
\end{aligned}$$

then we deduce that

$$\begin{aligned}
&(1+\xi)^{\alpha_1|\xi|} |T_1\psi_n - T_1\psi| \\
&= (1+\xi)^{\alpha_1|\xi|} \left| \int_{-\infty}^{\xi} e^{-\mu(\xi-\eta)} [h_1(\eta, \psi_n(\eta)) - h_1(\eta, \psi(\eta)) + \mu(\psi_n - \psi)] d\eta \right| \\
&\leq C(1+\xi)^{\alpha_1|\xi|} \left[e^{-\mu\xi} + \xi e^{-\frac{\mu}{2}\xi} + (1+\xi)^{-\frac{1}{1-m}} \right] \|\psi_n - \psi\|_X \\
&\leq C(1+\xi)^{-(\frac{1}{1-m}-\alpha_1)|\xi|} \|\psi_n - \psi\|_X.
\end{aligned}$$

If $f'(0) = s < f'(u_-)$, $k \geq 1$, the calculations are similar. When $0 < \eta < \frac{\xi}{2}$,

$$\begin{aligned}
&\left| \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} [h_1(\eta, \psi_n(\eta)) - h_1(\eta, \psi(\eta)) + \mu(\psi_n - \psi)] d\eta \right| \\
&\leq \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} |\psi_n - \psi| \left[(1+\eta)^{-1} + (1+\eta)^{-\frac{1-m}{k+1-m}} + (1+\eta)^{-\frac{k}{k+1-m}} \right. \\
&\quad \left. + e^{-(k+1-m)\Lambda_1|\eta|} + e^{-(1-m)\Lambda_1|\eta|} + 1 \right] d\eta \\
&\leq \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} (1+\eta)^{\frac{1}{k+1-m}} |\psi_n - \psi| (1+\eta)^{-\frac{1}{k+1-m}} d\eta \\
&\leq C e^{-\frac{\mu}{2}\xi} \|\psi_n - \psi\|_X \int_0^{\frac{\xi}{2}} (1+\eta)^{-\frac{1}{k+1-m}} d\eta \\
&\leq C \xi e^{-\frac{\mu}{2}\xi} \|\psi_n - \psi\|_X,
\end{aligned}$$

when $\frac{\xi}{2} < \eta < \xi$,

$$\begin{aligned}
&\left| \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} [h_1(\eta, \psi_n(\eta)) - h_1(\eta, \psi(\eta)) + \mu(\psi_n - \psi)] d\eta \right| \\
&\leq \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} |\psi_n - \psi| \left[(1+\eta)^{-1} + (1+\eta)^{-\frac{1-m}{k+1-m}} + (1+\eta)^{-\frac{k}{k+1-m}} \right. \\
&\quad \left. + e^{-(k+1-m)\Lambda_1|\eta|} + e^{-(1-m)\Lambda_1|\eta|} + 1 \right] d\eta \\
&\leq \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} (1+\eta)^{\frac{1}{k+1-m}} |\psi_n - \psi| (1+\eta)^{-\frac{1}{k+1-m}} d\eta \\
&\leq C(1+\xi)^{-\frac{1}{k+1-m}} \|\psi_n - \psi\|_X \int_{\frac{\xi}{2}}^{\xi} e^{-\mu(\xi-\eta)} d\eta \\
&\leq C(1+\xi)^{-\frac{1}{k+1-m}} \|\psi_n - \psi\|_X.
\end{aligned}$$

Hence, we can deduce that

$$(1+\xi)^{\alpha_1|\xi|} |T_1\psi_n - T_1\psi|$$

$$\begin{aligned}
&= (1 + \xi)^{\alpha_1 |\xi|} \left| \int_{-\infty}^{\xi} e^{-\mu(\xi-\eta)} [h_1(\eta, \psi_n(\eta)) - h_1(\eta, \psi(\eta)) + \mu(\psi_n - \psi)] d\eta \right| \\
&\leq C(1 + \xi)^{\alpha_1 |\xi|} \left[e^{-\mu\xi} + \xi e^{-\frac{\mu}{2}\xi} + (1 + \xi)^{-\frac{1}{k+1-m}} \right] \|\psi_n - \psi\|_X \\
&\leq C(1 + \xi)^{-(\frac{1}{k+1-m} - \alpha_1)|\xi|} \|\psi_n - \psi\|_X.
\end{aligned}$$

From the above calculations, $\forall \psi_n, \psi \in X$, $\|\psi_n - \psi\|_X \rightarrow 0$ ($n \rightarrow \infty$), we can obtain that

$$\|T_1 \psi_n - T_1 \psi\|_X \leq C \|\psi_n - \psi\|_X,$$

which implies that T_1 is continuous.

Letting $B_{R_X} = \{\psi \in X \mid \|\psi\|_X \leq R\}$, B_{R_X} is the bounded set in X . In view of (4.19), (4.20) and (4.21), $\forall \psi \in B_{R_X}$,

$$e^{\beta_1 |\xi|} |T_1 \psi| \leq C e^{-(\lambda_1 - \beta_1) |\xi|} \leq C, \quad \xi \leq 0, \quad (4.25)$$

when $\xi \geq 0$,

$$(1 + \xi)^{\alpha_1} |T_1 \psi| \leq \begin{cases} (1 + \xi)^{-(\frac{1}{1-m} - \alpha_1)} \leq C, & f'(0) < s \leq f'(u_-), \\ (1 + \xi)^{-(\frac{1}{k+1-m} - \alpha_1)} \leq C, & f'(0) = s < f'(u_-), \end{cases} \quad (4.26)$$

then we deduce that $\|T_1 \psi\| \leq C$, thus all functions in $T_1(B_{R_X})$ are uniformly bounded. From previous calculations, we already know that $h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)$ is bounded either $\xi \leq 0$ or $\xi \geq 0$. Choose $M_1 > 0$ such that

$$h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta) \leq M_1, \quad \forall \xi \in (-\infty, +\infty).$$

$\forall \varepsilon > 0$, $\exists \delta = \frac{\varepsilon}{2M_1} > 0$, $\forall \xi_1, \xi_2 \in \mathbb{R}$. Without loss of generality, suppose that $\xi_1 < \xi_2$,

$$\begin{aligned}
&|T_1 \psi(\xi_1) - T_1 \psi(\xi_2)| \\
&= \left| \int_{-\infty}^{\xi_1} e^{-\mu(\xi_1-\eta)} [h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)] d\eta \right. \\
&\quad \left. - \int_{-\infty}^{\xi_2} e^{-\mu(\xi_2-\eta)} [h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)] d\eta \right| \\
&= \int_{-\infty}^{\xi_1} |e^{-\mu(\xi_1-\eta)} - e^{-\mu(\xi_2-\eta)}| |h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)| d\eta \\
&\quad + \int_{\xi_1}^{\xi_2} e^{-\mu(\xi_2-\eta)} |h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)| d\eta \\
&\triangleq I_1 + I_2.
\end{aligned}$$

Employing the Mean Value Theorem, we have

$$\begin{aligned}
I_1 &= \int_{-\infty}^{\xi_1} |-\mu e^{-\mu(\gamma-\eta)} (\xi_1 - \xi_2)| |h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)| d\eta \\
&\leq M_1 \mu |\xi_1 - \xi_2| \int_{-\infty}^{\xi_1} e^{-\mu(\gamma-\eta)} d\eta \\
&\leq M_1 \mu |\xi_1 - \xi_2| \int_{-\infty}^{\xi_1} e^{-\mu(\xi_1-\eta)} d\eta
\end{aligned}$$

$$= M_1 |\xi_1 - \xi_2|,$$

where γ between ξ_1 and ξ_2 . Utilizing Taylor's expansion, we can see that

$$\begin{aligned} I_2 &= \int_{\xi_1}^{\xi_2} e^{-\mu(\xi_2-\eta)} |h_1(\eta, \psi(\eta)) - \ell_{1\eta} + \mu\psi(\eta)| \, d\eta \\ &\leq M_1 \int_{\xi_1}^{\xi_2} e^{-\mu(\xi_2-\eta)} \, d\eta \\ &= \frac{M_1}{\mu} (1 - e^{-\mu(\xi_2-\xi_1)}) \\ &= \frac{M_1}{\mu} [1 - (1 - \mu(\xi_2 - \xi_1) + o((\xi_2 - \xi_1)^2))] \\ &\leq M_1 |\xi_1 - \xi_2|. \end{aligned}$$

$\forall \psi \in B_{R_X}$, when $|\xi_1 - \xi_2| < \delta$,

$$|T_1\psi(\xi_1) - T_1\psi(\xi_2)| \leq 2M_1 |\xi_1 - \xi_2| \leq 2M_1 \delta < \varepsilon,$$

which implies that all functions in $T_1(B_{R_X})$ are equicontinuous. Reviewing (4.25) and (4.26), we have

$$\lim_{\xi \rightarrow -\infty} e^{\beta_1|\xi|} |T_1\psi(\xi)| = 0,$$

and

$$\lim_{\xi \rightarrow +\infty} (1 + \xi)^{\alpha_1} |T_1\psi(\xi)| = 0.$$

To sum up, $T_1 : X \rightarrow X$ is completely continuous.

$\forall \psi \in X$, $\psi(\xi) = \theta T_1\psi(\xi)$, $0 < \theta < 1$, when $\xi \leq 0$,

$$e^{\beta_1|\xi|} |\psi(\xi)| = e^{\beta_1|\xi|} \theta |T_1\psi(\xi)| \leq e^{\beta_1|\xi|} |T_1\psi(\xi)| \leq C e^{-(\Lambda_1 - \beta_1)|\xi|} \leq C,$$

when $\xi \geq 0$, if $f'(0) < s \leq f'(u_-)$,

$$(1 + \xi)^{\alpha_1} |\psi(\xi)| = (1 + \xi)^{\alpha_1} \theta |T_1\psi(\xi)| \leq (1 + \xi)^{\alpha_1} |T_1\psi(\xi)| \leq C(1 + \xi)^{-(\frac{1}{1-m} - \alpha_1)} \leq C,$$

or if $f'(0) = s < f'(u_-)$,

$$(1 + \xi)^{\alpha_1} |\psi(\xi)| = (1 + \xi)^{\alpha_1} \theta |T_1\psi(\xi)| \leq (1 + \xi)^{\alpha_1} |T_1\psi(\xi)| \leq C(1 + \xi)^{-(\frac{1}{k+1-m} - \alpha_1)} \leq C,$$

then we obtain that

$$\|\psi\|_X \leq C.$$

There exists a solution to the equation (4.3) in the closed sphere $B(0, R_1)$ of X by employing Leray-Schauder theorem, where

$$R_1 = \sup \{ \|\psi\|_X \mid \psi = \theta T_1\psi, 0 < \theta < 1 \}.$$

Let ϕ be the travelling wave solution obtained in X . Since ϕ satisfies equations (2.1) and connects the constant states u_- and 0, and all hypotheses of Proposition 3.1 are ensured, the analysis in Proposition 3.1 yields precisely the algebraic decay rates as $\xi \rightarrow +\infty$ and the exponential decay (monotonic or oscillatory) as $\xi \rightarrow -\infty$ as stated in the theorem. The proof of Theorem 2.1 is complete. $\square$

Proof of Theorem 2.2. For $m = 0$, select a smooth function

$$\ell_2(\xi) = u_- - u_- \frac{1}{1 + e^{-\Lambda_2 \xi}},$$

which satisfies $\ell_2(\pm\infty) = u_{\pm}$, where $\Lambda_2 = \min\{\lambda_2, \nu_2\} > 0$. Define $\psi = \phi - \ell_2$, then we have

$$\begin{cases} \psi_{\xi} = \frac{1}{D}(\psi + \ell_2) [f((\psi + \ell_2)(\xi + sr)) - f(u_{\pm}) - s(\psi + \ell_2 - u_{\pm})] - \ell_{2\xi}, \\ \psi(\pm\infty) = 0, \end{cases} \quad (4.27)$$

the equation of ψ can be deduced by equation (3.24) and $\phi(\pm\infty) = u_{\pm}$. Similar to $0 < m < 1$, we know that

$$\begin{cases} |\ell_2(\xi)| \leq Ce^{-\Lambda_2|\xi|}, & \xi \geq 0, \\ |\ell_2(\xi) - u_-| \leq Ce^{-\Lambda_2|\xi|}, & \xi \leq 0, \\ |\ell_{2\xi}| \leq Ce^{-\Lambda_2|\xi|}, & \xi \in (-\infty, +\infty), \end{cases} \quad \begin{cases} |\psi| \leq Ce^{-\Lambda_2|\xi|}, & \xi \rightarrow -\infty, \\ |\psi| \leq C\xi^{-1}, & \xi \rightarrow +\infty, f'(0) < s \leq f'(u_-), \\ |\psi| \leq C\xi^{-\frac{1}{k+1}}, & \xi \rightarrow +\infty, f'(0) = s < f'(u_-). \end{cases}$$

We define a space

$$Y = \left\{ \psi(\xi) \in C(\mathbb{R}) \mid \|\psi\|_Y < \infty, \psi \sim C_+ \xi^{-\frac{1}{k+1}} (\xi \rightarrow +\infty), \psi \sim C_- e^{-\Lambda_2|\xi|} (\xi \rightarrow -\infty) \right\},$$

and the weighted norm

$$\|\psi\|_Y = \max \left\{ \sup_{\xi \geq 0} (1 + \xi)^{\alpha_2} |\psi(\xi)|, \sup_{\xi \leq 0} e^{\beta_2|\xi|} |\psi(\xi)| \right\},$$

where $0 < \alpha_2 < \frac{1}{k+1}$ and $0 < \beta_2 < \Lambda_2$.

Denote $\psi_{\xi} \triangleq h_2(\xi, \psi(\xi)) - \ell_{2\xi}$, consider the integral equation

$$\psi(\xi) = \int_{-\infty}^{\xi} e^{-\mu(\xi-\eta)} [h_2(\eta, \psi(\eta)) - \ell_{2\eta} + \mu\psi(\eta)] d\eta, \quad (4.28)$$

we will obtain

$$\begin{aligned} |\psi(\xi)| &\leq Ce^{-\mu\xi} + C\xi(e^{-\frac{\mu}{2}\xi} + e^{-\frac{\Lambda_2}{2}\xi}) + C(1 + \xi)^{-1}, \quad f'(0) < s \leq f'(u_-), \quad \xi \geq 0, \\ |\psi(\xi)| &\leq Ce^{-\mu\xi} + C\xi(e^{-\frac{\mu}{2}\xi} + e^{-\frac{\Lambda_2}{2}\xi}) + C(1 + \xi)^{-\frac{1}{k+1}}, \quad f'(0) = s < f'(u_-), \quad \xi \geq 0, \end{aligned}$$

and

$$|\psi(\xi)| \leq Ce^{-\Lambda_2|\xi|}, \quad \xi \leq 0,$$

which can be obtained by the estimation of $h_2(\eta, \psi(\eta)) - \ell_{2\eta} + \mu\psi(\eta)$ later, thus equation (4.28) is equivalent to the integral equation (4.27). Define

$$T_2\psi(\xi) = \int_{-\infty}^{\xi} e^{-\mu(\xi-\eta)} [h_2(\eta, \psi(\eta)) - \ell_{2\eta} + \mu\psi(\eta)] d\eta.$$

$\forall \psi \in Y$, we can see that

$$\begin{cases} |\psi| \leq Ce^{-\Lambda_2|\xi|}, & \xi \rightarrow -\infty, \\ |\psi| \leq C(1 + \xi)^{-1}, & \xi \rightarrow +\infty, f'(0) < s \leq f'(u_-), \\ |\psi| \leq C(1 + \xi)^{-\frac{1}{k+1}}, & \xi \rightarrow +\infty, f'(0) = s < f'(u_-). \end{cases}$$

When $\eta \leq 0$,

$$\begin{aligned} |h_2(\eta, \psi(\eta))| &= \frac{1}{D} \left| (\psi + \ell_2)(\psi + \ell_2 - u_-) \left[\frac{f(\psi + \ell_2)(\eta + sr) - f(u_-)}{\psi + \ell_2 - u_-} - s \right] \right| \\ &\leq C(|\psi|^2 + |\psi||\ell_2 - u_-| + \ell_2|\psi| + \ell_2|\ell_2 - u_-|) \\ &\leq Ce^{-\Lambda_2|\eta|}. \end{aligned}$$

When $\eta \geq 0$, if $f'(0) < s \leq f'(u_-)$, $k = 0$,

$$\begin{aligned} |h_2(\eta, \psi(\eta))| &= \frac{1}{D} \left| (\psi + \ell_2)^2 \left[\frac{f(\psi + \ell_2)(\eta + sr) - f(0)}{\psi + \ell_2} - s \right] \right| \\ &\leq C(|\psi|^2 + |\ell_2|^2) \\ &\leq C(1 + \eta)^{-2} + e^{-2\Lambda_2|\eta|}, \end{aligned}$$

or if $f'(0) = s < f'(u_-)$, $k \geq 1$,

$$\begin{aligned} |h_2(\eta, \psi(\eta))| &= \frac{1}{D} \left| (\psi + \ell_2) \left[f'(0)(\psi + \ell_2)(\eta + sr) - s(\psi + \ell_2)(\eta) \right. \right. \\ &\quad \left. \left. + \frac{f^{(k+1)}(0)}{(k+1)!}(\psi + \ell_2)^{k+1}(\eta + sr) + O((\psi + \ell_2)^{k+2}(\eta + sr)) \right] \right| \\ &\leq C|\psi + \ell_2|^2 + C|\psi + \ell_2|^{k+2} \\ &\leq C(|\psi|^2 + |\psi|^{k+2} + |\ell_2|^2 + |\ell_2|^{k+2}) \\ &\leq C(1 + \eta)^{-\frac{2}{k+1}} + C(1 + \eta)^{-1-\frac{1}{k+1}} + e^{-2\Lambda_2|\eta|} + e^{-(k+2)\Lambda_2|\eta|}. \end{aligned}$$

Base on the estimation of $|h_2(\eta, \psi(\eta))|$, similar to $0 < m < 1$, we can obtain the following estimations about $T_2\psi(\xi)$

$$\begin{cases} |T_2\psi(\xi)| \leq Ce^{-\Lambda_2|\xi|}, & \xi \leq 0, \\ |T_2\psi(\xi)| \leq Ce^{-\mu\xi} + C\xi(e^{-\frac{\mu}{2}\xi} + e^{-\frac{\Lambda_2}{2}\xi}) + C(1 + \xi)^{-1}, & \xi \geq 0, f'(0) < s \leq f'(u_-), \\ |T_2\psi(\xi)| \leq Ce^{-\mu\xi} + C\xi(e^{-\frac{\mu}{2}\xi} + e^{-\frac{\Lambda_2}{2}\xi}) + C(1 + \xi)^{-\frac{1}{k+1}}, & \xi \geq 0, f'(0) = s < f'(u_-), \end{cases}$$

then we deduce that

$$\begin{cases} e^{\beta_2|\xi|}|T_2\psi(\xi)| \leq Ce^{-(\Lambda_2-\beta_2)|\xi|}, & \xi \leq 0, \\ (1 + \xi)^{\alpha_2}|T_2\psi(\xi)| \leq C(1 + \xi)^{-(1-\alpha_2)}, & \xi \geq 0, f'(0) < s \leq f'(u_-), \\ (1 + \xi)^{\alpha_2}|T_2\psi(\xi)| \leq C(1 + \xi)^{-(\frac{1}{k+1}-\alpha_2)}, & \xi \geq 0, f'(0) = s < f'(u_-), \end{cases} \quad (4.29)$$

thus $\|T_2\psi(\xi)\|_Y \leq C$, and $T_2\psi$ converges exponentially to 0 at around $-\infty$ and converges algebraically to 0 at around $+\infty$, which implies that $T_2\psi \in Y$.

Letting $\psi_n, \psi \in Y$, $\|\psi_n - \psi\|_Y \rightarrow 0$ ($n \rightarrow \infty$),

$$|T_2\psi_n - T_2\psi| = \left| \int_{-\infty}^{\xi} e^{-\mu(\xi-\eta)} [h_2(\eta, \psi_n(\eta)) - h_2(\eta, \psi(\eta)) + \mu(\psi_n - \psi)] d\eta \right|.$$

We are going to estimate the error term, when $\eta \leq 0$,

$$|h_2(\eta, \psi_n(\eta)) - h_2(\eta, \psi(\eta))|$$

$$\begin{aligned}
&\leq \frac{1}{D} \left| [(\psi_n + \ell_2) - (\psi + \ell_2)] (\psi_n + \ell_2 - u_-) \left[\frac{f(\psi_n + \ell_2)(\eta + sr) - f(u_-)}{\psi_n + \ell_2 - u_-} - s \right] \right. \\
&\quad \left. + \frac{1}{D} \left| (\psi + \ell_2)(\psi_n - \psi) \left[\frac{(f(\psi_n + \ell_2) - f(\psi + \ell_2))(\eta + sr)}{(\psi_n + \ell_2) - (\psi + \ell_2)} - s \right] \right| \right| \\
&\leq C|\psi_n - \psi| (|\psi_n| + |\ell_2 - u_-| + |\psi| + |\ell_2|) \\
&\leq C|\psi_n - \psi|(e^{-\Lambda_2|\eta|} + 1).
\end{aligned}$$

When $\eta \geq 0$, if $f'(0) < s \leq f'(u_-)$, $k = 0$, then

$$\begin{aligned}
&|h_2(\eta, \psi_n(\eta)) - h_2(\eta, \psi(\eta))| \\
&\leq \frac{1}{D} \left| [(\psi_n + \ell_2) - (\psi + \ell_2)] (\psi_n + \ell_2) \left[\frac{f(\psi_n + \ell_2)(\eta + sr) - f(0)}{\psi_n + \ell_2 - 0} - s \right] \right. \\
&\quad \left. + \frac{1}{D} \left| (\psi + \ell_2)(\psi_n - \psi) \left[\frac{(f(\psi_n + \ell_2) - f(\psi + \ell_2))(\eta + sr)}{(\psi_n + \ell_2) - (\psi + \ell_2)} - s \right] \right| \right| \\
&\leq C|(\psi_n - \psi)(\psi_n + \ell_2)| + C|(\psi_n - \psi)(\psi + \ell_2)| \\
&\leq C|\psi_n - \psi| (|\psi_n| + |\psi| + |\ell_2|) \\
&\leq C|\psi_n - \psi| \left[(1 + \eta)^{-1} + e^{-\Lambda_2|\eta|} \right],
\end{aligned}$$

or if $f'(0) = s < f'(u_-)$, $k \geq 1$,

$$\begin{aligned}
&|h_2(\eta, \psi_n(\eta)) - h_2(\eta, \psi(\eta))| \\
&\leq \frac{1}{D} \left| (\psi_n - \psi) \left[f'(0)(\psi_n + \ell_2)(\eta + sr) - s(\psi_n + \ell_2) \right. \right. \\
&\quad \left. \left. + \frac{f^{(k+1)}(0)}{(k+1)!} (\psi_n + \ell_2)^{k+1} (\eta + sr) + o((\psi_n + \ell_2)^{k+1}) \right] \right| \\
&\quad + \frac{1}{D} \left| (\psi + \ell_2) \left[f'(0)(\psi_n - \psi)(\eta + sr) - s(\psi_n - \psi) \right. \right. \\
&\quad \left. \left. + \frac{f^{(k+1)}(0)}{(k+1)!} (\psi_n^{k+1}(\eta + sr) - \psi^{k+1}(\eta + sr)) + o(\psi_n^{k+2}(\eta + sr)) + o(\psi^{k+2}(\eta + sr)) \right] \right| \\
&\leq C|\psi_n - \psi| (|\psi_n + \ell_2| + |\psi_n + \ell_2|^{k+1}(\eta + sr)) \\
&\quad + C(|\psi| + |\ell_2|) (|\psi_n - \psi| + |\psi_n^{k+1} - \psi^{k+1}|(\eta + sr)) \\
&\leq C|\psi_n - \psi| (|\psi_n| + |\psi_n|^{k+1} + |\psi| + |\psi_n - \psi|^k + |\ell_2| + |\ell_2|^{k+1}) \\
&\leq C|\psi_n - \psi| \left[(1 + \eta)^{-1} + (1 + \eta)^{-\frac{k}{k+1}} + (1 + \eta)^{-\frac{1}{k+1}} + e^{-(k+1)\Lambda_2|\eta|} + e^{-\Lambda_2|\eta|} \right].
\end{aligned}$$

Thanks to the estimation of $|h_2(\eta, \psi_n(\eta)) - h_2(\eta, \psi(\eta))|$, we can calculate that

$$\begin{cases} e^{\beta_2|\xi|} |T_2\psi_n - T_2\psi| \leq C e^{-(\Lambda_2 - \beta_2)|\xi|} \|\psi_n - \psi\|_Y, & \xi \leq 0, \\ (1 + \xi)^{\alpha_2} |T_2\psi_n - T_2\psi| \leq C(1 + \xi)^{-(1-\alpha_2)} \|\psi_n - \psi\|_Y, & \xi \geq 0, f'(0) < s \leq f'(u_-), \\ (1 + \xi)^{\alpha_2} |T_2\psi_n - T_2\psi| \leq C(1 + \xi)^{-(\frac{1}{k+1} - \alpha_2)} \|\psi_n - \psi\|_Y, & \xi \geq 0, f'(0) = s < f'(u_-), \end{cases} \quad (4.30)$$

thus

$$\|T_2\psi_n - T_2\psi\|_Y \leq C \|\psi_n - \psi\|_Y,$$

which implies that T_2 is continuous.

Letting $B_{R_Y} = \{\psi \in Y \mid \|\psi\|_Y \leq R\}$, B_{R_Y} is the bounded set in Y , and it is easy to see that $\|T_2\psi\|_Y \leq C$ due to (4.30), thus all functions in $T_2(B_{R_Y})$ are uniformly bounded. Since $h_2(\eta, \psi(\eta)) - \ell_{2\eta} + \mu\psi(\eta)$ is bounded, select $M_2 > 0$ such that

$$[h_2(\eta, \psi(\eta)) - \ell_{2\eta} + \mu\psi(\eta)] \leq M_2, \quad \forall \xi \in (-\infty, +\infty).$$

$\forall \varepsilon > 0$, $\exists \delta = \frac{\varepsilon}{M_2(1+\mu)} > 0$, $\forall \xi_1, \xi_2 \in \mathbb{R}$, $\forall \psi \in B_{R_Y}$, similar to the estimation of $0 < m < 1$, when $|\xi_1 - \xi_2| < \delta$,

$$|T_2\psi(\xi_1) - T_2\psi(\xi_2)| \leq M_2(1+\mu)|\xi_1 - \xi_2| \leq M_2(1+\mu)\delta < \varepsilon.$$

In view of (4.29),

$$\lim_{\xi \rightarrow -\infty} e^{\beta_2|\xi|} |T_2\psi(\xi)| = 0,$$

and

$$\lim_{\xi \rightarrow +\infty} (1+\xi)^{\alpha_2} |T_2\psi(\xi)| = 0,$$

then $T_2 : Y \rightarrow Y$ is completely continuous.

$\forall \psi \in Y$, $\psi(\xi) = \theta T_2\psi(\xi)$, $0 < \theta < 1$. According to (4.29),

$$\begin{cases} e^{\beta_2|\xi|} |\psi(\xi)| = e^{\beta_2|\xi|} \theta |T_2\psi(\xi)| \leq C e^{-(\Lambda_2 - \beta_2)|\xi|} \leq C, & \xi \leq 0, \\ (1+\xi)^{\alpha_2} |\psi(\xi)| \leq C(1+\xi)^{-(1-\alpha_2)} \theta |T_2\psi(\xi)| \leq C, & \xi \geq 0, \quad f'(0) < s \leq f'(u_-), \\ (1+\xi)^{\alpha_2} |\psi(\xi)| \leq C(1+\xi)^{-(\frac{1}{k+1}-\alpha_2)} \theta |T_2\psi(\xi)| \leq C, & \xi \geq 0, \quad f'(0) = s < f'(u_-), \end{cases}$$

then we obtain that

$$\|\psi\|_Y \leq C.$$

There exists a solution to the equation (??) in the closed sphere $B(0, R_2)$ of Y by employing Leray-Schauder theorem, where

$$R_2 = \sup \{ \|\psi\|_Y \mid \psi = \theta T_2\psi, 0 < \theta < 1 \}.$$

The proof of the asymptotic behavior follows readily from Proposition 3.2. Thus the proof of Theorem 2.2 is complete. $\square$

Proof of Theorem 2.3. For $m < 0$, select a smooth function

$$\ell_3(\xi) = u_- - u_- \frac{1}{1 + e^{-\Lambda_3 \xi}},$$

which satisfies $\ell_3(\pm\infty) = u_{\pm}$ and $\Lambda_3 = \min\{\lambda_3, \nu_3\} > 0$. Define $\psi = \phi - \ell_3$, then we have

$$\begin{cases} \psi_\xi = \frac{1}{D}(\psi + \ell_3)^{1+|m|} [f((\psi + \ell_3)(\xi + sr)) - f(u_{\pm}) - s(\psi + \ell_3 - u_{\pm})] - \ell_{3\xi}, \\ \psi(\pm\infty) = 0, \end{cases} \quad (4.31)$$

the equation of ψ can be deduced by equation (3.28) and $\phi(\pm\infty) = u_{\pm}$. Similar to previous derivation, we know that

$$\begin{cases} |\ell_3(\xi)| \leq C e^{-\Lambda_3|\xi|}, & \xi \geq 0, \\ |\ell_3(\xi) - u_-| \leq C e^{-\Lambda_3|\xi|}, & \xi \leq 0, \\ |\ell_{3\xi}| \leq C e^{-\Lambda_3|\xi|}, & \xi \in (-\infty, +\infty), \end{cases}$$

and

$$\begin{cases} |\psi| \leq C e^{-\Lambda_3 |\xi|}, & \xi \rightarrow -\infty, \\ |\psi| \leq C \xi^{-\frac{1}{1+m}}, & \xi \rightarrow +\infty, f'(0) < s \leq f'(u_-), \\ |\psi| \leq C \xi^{-\frac{1}{k+1+m}}, & \xi \rightarrow +\infty, f'(0) = s < f'(u_-). \end{cases}$$

We define a space

$$Z = \left\{ \psi(\xi) \in C(\mathbb{R}) \mid \|\psi\|_Z < \infty, \psi \sim C_+ \xi^{-\frac{1}{k+1+m}} (\xi \rightarrow +\infty), \psi \sim C_- e^{-\Lambda_3 |\xi|} (\xi \rightarrow -\infty) \right\},$$

and the weighted norm

$$\|\psi\|_Z = \max \left\{ \sup_{\xi \geq 0} (1 + \xi)^{\alpha_3} |\psi(\xi)|, \sup_{\xi \leq 0} e^{\beta_3 |\xi|} |\psi(\xi)| \right\},$$

where $0 < \alpha_3 < \frac{1}{k+1+m}$ and $0 < \beta_3 < \Lambda_3$. There are only minor differences between $0 < m < 1$ and $m < 0$, here we have omitted the details. $\square$

5 Numerical simulations

This section presents numerical solutions of (2.1)–(2.3) corresponding to the various cases in Theorems 2.1–2.3. Throughout, we set $D = 1$, $u_- = 1$, $u_+ = 0$, and consider the flux functions

$$f_1(u) = (u - 1)^2, \quad f_2(u) = -u^3 + 2u^2 - \frac{1}{2}u, \quad f_3(u) = \frac{1}{2}u^2.$$

One readily verifies that

- $f_1(u_-) = 0$, $f_1(u_+) = 1$, $s_1 = -1$, $f'_1(u_-) = 0$, and $s < 0 = f'_1(u_-)$.
- $f_2(u_-) = \frac{1}{2}$, $f_2(u_+) = 0$, $s_2 = \frac{1}{2}$, $f'_2(u_-) = \frac{1}{2}$, and $0 < s = f'_2(u_-)$.
- $f_3(u_-) = \frac{1}{2}$, $f_3(u_+) = 0$, $s_3 = \frac{1}{2}$, $f'_3(u_-) = 1$, and $0 < s < f'_3(u_-)$.

Our computational approach employs domain decomposition in time: we partition a sufficiently large time interval into subintervals of length equal to the delay, then solve sequentially using the Crank–Nicolson scheme combined with Newton's iteration. Our numerical results are list as follows.

Case $0 < m < 1$: We fix $m = \frac{1}{2}$ and adopt the initial data depending on a in the form of

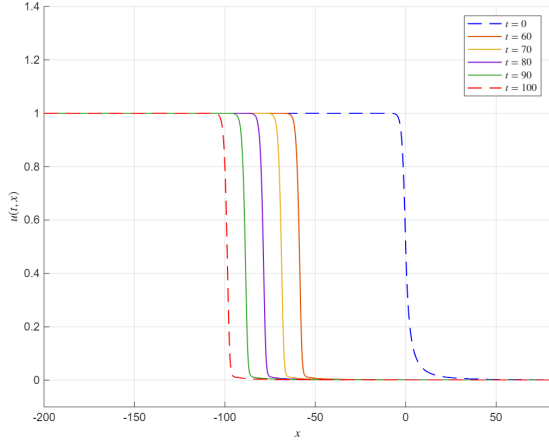
$$u_a^{(1)}(t, x) = \begin{cases} \frac{1}{1 + e^{ax}}, & \text{for } t \leq 0, x \leq 0 \\ \frac{1}{(4 + ax)^2}, & \text{for } t \leq 0, x > 0 \end{cases} \quad (5.1)$$

which is C^1 on $\mathbb{R}^- \times \mathbb{R}$ and captures various asymptotic decay rates depending on $a > 0$.

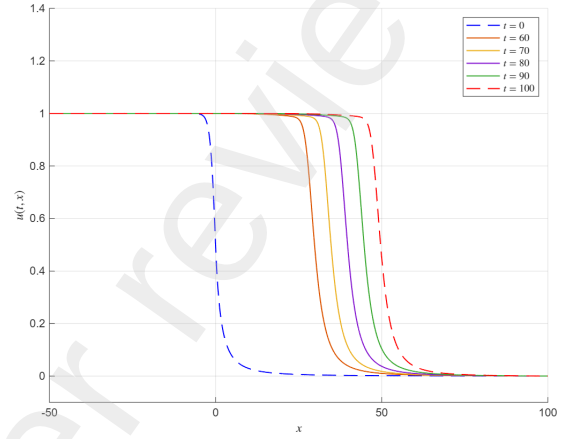
(1.1) Let $f = f_1$, $r = 1$, and $u_0(\tau, x) = u_{\lambda_1}^{(1)}(\tau, x)$ for $\tau \in [-r, 0]$, where $\lambda_1 = 1.0$ is the unique real root of (2.6). The computational domain is $[-200, 80]$. Figure 1(a) displays a monotone traveling wave.

(1.2) Let $f = f_2$, $r = 2 < \hat{r}_1$, and $u_0(\tau, x) = u_{\lambda_1}^{(1)}(\tau, x)$ for $\tau \in [-r, 0]$, where $\lambda_1 \approx 1.25643$ is the smallest real root of (2.6). The computational domain is $[-50, 100]$. Figure 1(b) displays a monotone traveling wave.

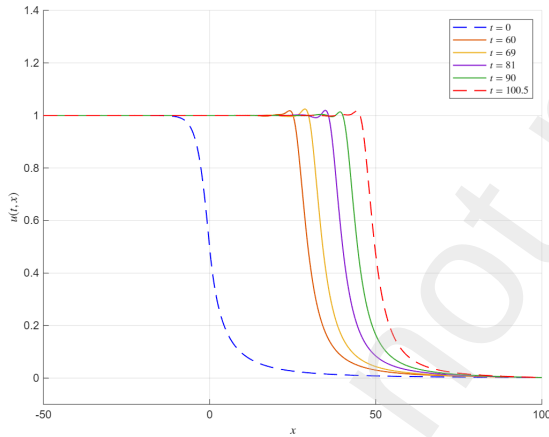
- (1.3) Let $f = f_3$, $r = 1.5 < \bar{r}_1^{(1)}$, and $u_0(\tau, x) = u_{v_1}^{(1)}(\tau, x)$ for $\tau \in [-r, 0]$, where $v_1 \approx 0.53122$ approximates $\Lambda_1^{(1)}$ from Theorem 2.1. The computational domain is $[-50, 100]$. Figure 1(c) exhibits oscillations behavior.
- (1.4) Let $f = f_2$, $r = 5.0 > \hat{r}_1$, and $u_0(\tau, x) = u_{v_1}^{(1)}(\tau, x)$ for $\tau \in [-r, 0]$, where $v_1 \approx 0.74666$ approximates $\Lambda_1^{(2)}$ from Theorem 2.1. The computational domain is $[-50, 100]$. Figure 1(d) exhibits oscillations behavior.



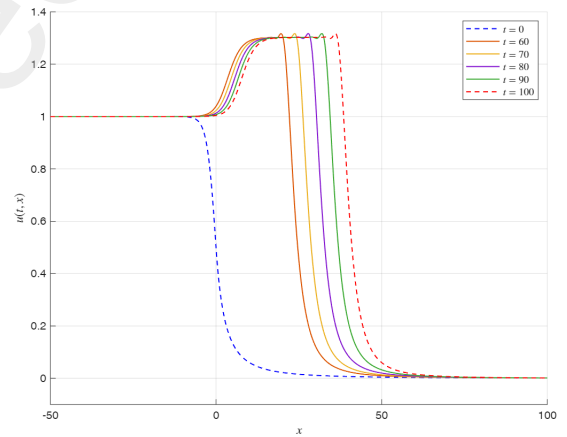
(a) $f_1(u) = (u - 1)^2$, $r = 1$



(b) $f_2(u) = -u^3 + 2u^2 - \frac{1}{2}u$, $r = 2$



(c) $f_3(u) = \frac{1}{2}u^2$, $r = 1.5$



(d) $f_2(u) = -u^3 + 2u^2 - \frac{1}{2}u$, $r = 5$

Figure 1: Numerical solutions for $m = \frac{1}{2}$: monotone waves in (a)–(b); oscillations waves in (c)–(d).

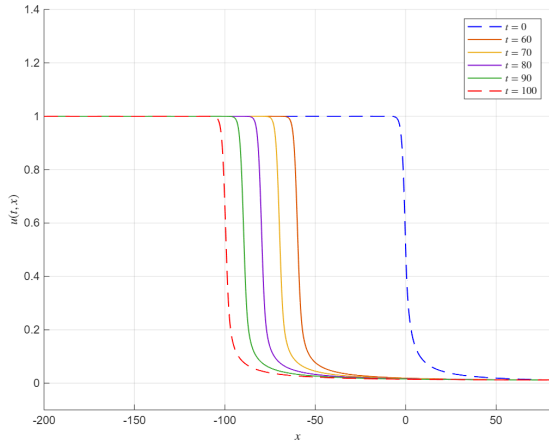
Case $m = 0$: We adopt the initial data

$$u_a^{(2)}(t, x) = \begin{cases} \frac{1}{1 + e^{ax}}, & \text{for } t \leq 0, x \leq 0 \\ \frac{1}{2 + ax}, & \text{for } t \leq 0, x > 0 \end{cases} \quad (5.2)$$

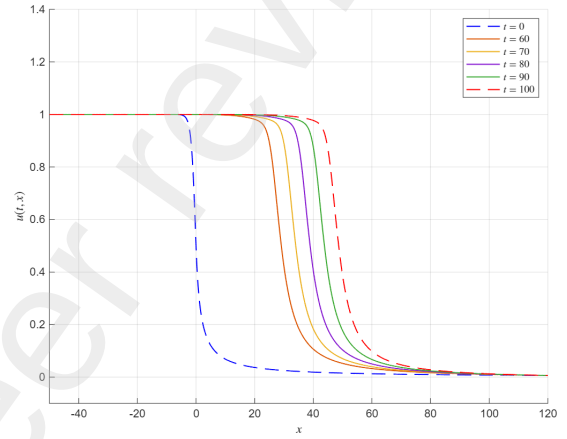
which is C^1 on $\mathbb{R}^- \times \mathbb{R}$.

- (2.1) Let $f = f_1$, $r = 1$, and $u_0(\tau, x) = u_{\lambda_2}^{(2)}(\tau, x)$ for $\tau \in [-r, 0]$, where $\lambda_2 = 1.0$ is the unique real root of (2.11). The computational domain is $[-200, 80]$. Figure 2(a) displays a monotone traveling wave.

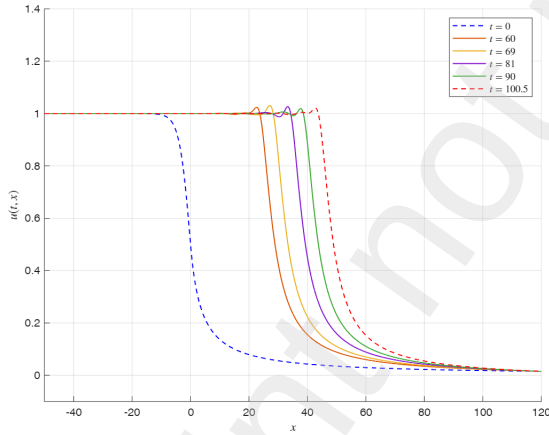
- (2.2) Let $f = f_2$, $r = 2 < \hat{r}_2$, and $u_0(\tau, x) = u_{\lambda_2}^{(2)}(\tau, x)$ for $\tau \in [-r, 0]$, where $\lambda_2 \approx 1.25643$ is the smallest real root of (2.11). The computational domain is $[-50, 120]$. Figure 2(b) displays a monotone traveling wave.
- (2.3) Let $f = f_3$, $r = 1.5 < \bar{r}_2^{(1)}$, and $u_0(\tau, x) = u_{\nu_2}^{(2)}(\tau, x)$ for $\tau \in [-r, 0]$, where $\nu_2 \approx 0.53122$ approximates $\Lambda_2^{(1)}$ from Theorem 2.2. The computational domain is $[-50, 120]$. The results are shown in Figure 2(c), illustrating the oscillations behavior.
- (2.4) Let $f = f_2$, $r = 5.0 > \hat{r}_2$, and $u_0(\tau, x) = u_{\nu_2}^{(2)}(\tau, x)$ for $\tau \in [-r, 0]$, where $\nu_2 \approx 0.74666$ approximates $\Lambda_2^{(2)}$ from Theorem 2.2. The computational domain is $[-50, 120]$. The results are shown in Figure 2(d), illustrating the oscillations behavior.



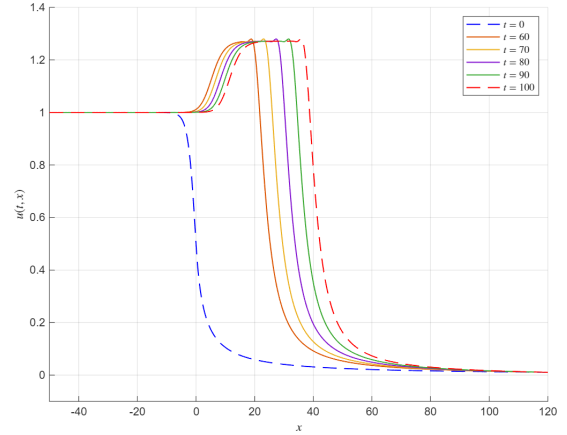
(a) $f_1(u) = (u - 1)^2$, $r = 1$



(b) $f_2(u) = -u^3 + 2u^2 - \frac{1}{2}u$, $r = 2$



(c) $f_3(u) = \frac{1}{2}u^2$, $r = 1.5$



(d) $f_2(u) = -u^3 + 2u^2 - \frac{1}{2}u$, $r = 5$

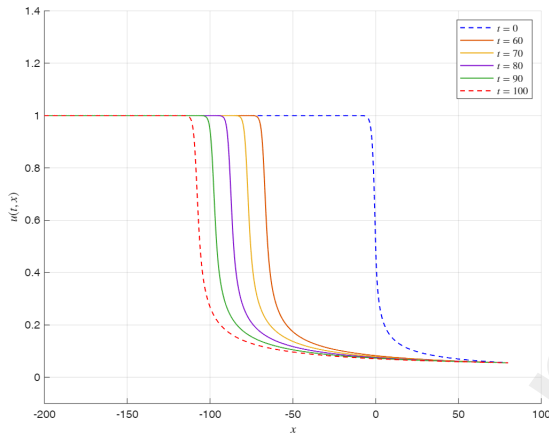
Figure 2: Numerical solutions for $m = 0$: monotone waves in (a)–(b); oscillations waves in (c)–(d).

Case $m < 0$: We fix $m = -1$ and adopt the initial data depending on a in the form of

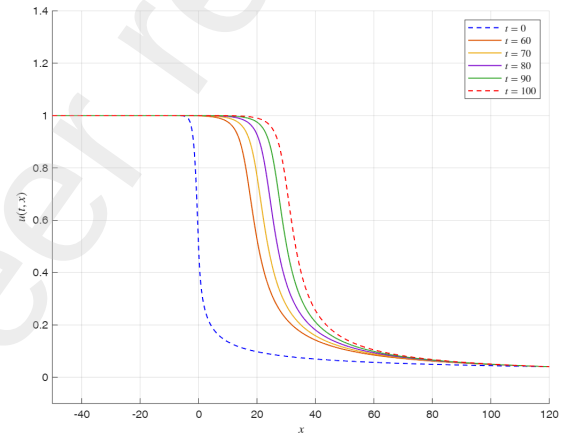
$$u_a^{(3)}(t, x) = \begin{cases} \frac{1}{1 + e^{ax}}, & \text{for } t \leq 0, x \leq 0 \\ \frac{1}{2\sqrt{1 + ax}}, & \text{for } t \leq 0, x > 0 \end{cases} \quad (5.3)$$

which is C^1 on $\mathbb{R}^- \times \mathbb{R}$.

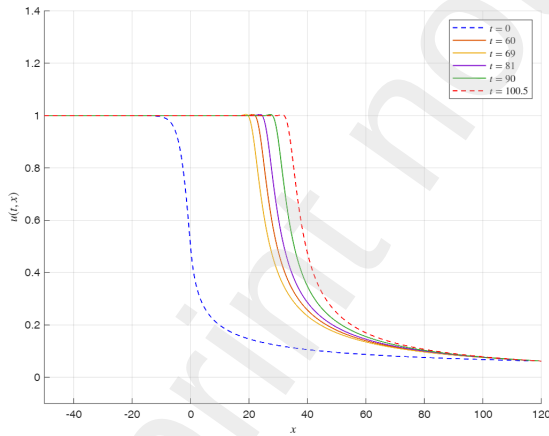
- (3.1) Let $f = f_1$, $r = 1$, and $u_0(\tau, x) = u_{\lambda_3}^{(3)}(\tau, x)$ for $\tau \in [-r, 0]$, where $\lambda_3 = 1.0$ is the unique real root of (2.16). The computational domain is $[-200, 80]$. Figure 3(a) displays a monotone traveling wave.
- (3.2) Let $f = f_2$, $r = 2 < \hat{r}_3$, and $u_0(\tau, x) = u_{\lambda_3}^{(3)}(\tau, x)$ for $\tau \in [-r, 0]$, where $\lambda_3 \approx 1.25643$ is the smallest real root of (2.16). The computational domain is $[-50, 120]$. Figure 3(b) displays a monotone traveling wave.
- (3.3) Let $f = f_3$, $r = 1.5 < \hat{r}_3^{(1)}$, and $u_0(\tau, x) = u_{\nu_3}^{(3)}(\tau, x)$ for $\tau \in [-r, 0]$, where $\nu_3 \approx 0.53122$ approximates $\Lambda_3^{(1)}$ from Theorem 2.3. The computational domain is $[-50, 120]$. Figure 3(c) exhibits oscillatory behavior.
- (3.4) Let $f = f_2$, $r = 5.0 > \hat{r}_3$, and $u_0(\tau, x) = u_{\nu_3}^{(3)}(\tau, x)$ for $\tau \in [-r, 0]$, where $\nu_3 \approx 0.74666$ approximates $\Lambda_3^{(2)}$ from Theorem 2.3. The computational domain is $[-50, 120]$. Figure 3(d) exhibits oscillatory behavior.



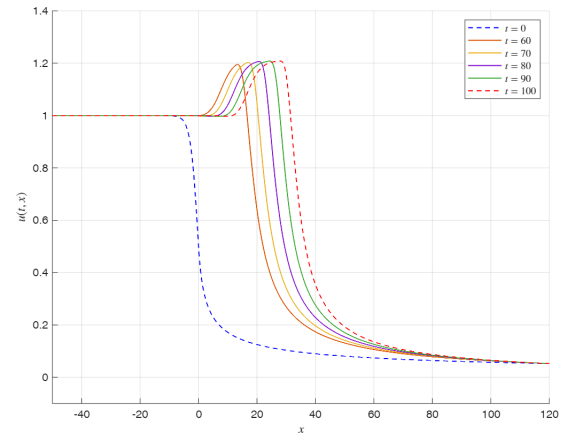
(a) $f_1(u) = (u - 1)^2$, $r = 1$



(b) $f_2(u) = -u^3 + 2u^2 - \frac{1}{2}u$, $r = 2$



(c) $f_3(u) = \frac{1}{2}u^2$, $r = 1.5$



(d) $f_2(u) = -u^3 + 2u^2 - \frac{1}{2}u$, $r = 5$

Figure 3: Numerical solutions for $m = -1$: monotone waves in (a)–(b); oscillations waves in (c)–(d).

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